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ORGANIC LIGHT EMITTING DIODE AND ORGANIC LIGHT EMITTING DEVICE INCLUDING THE SAME

Abstract

Organic light-emitting diodes and organic light emitting devices including the same are described. The organic light emitting diode can include a first electrode; a second electrode facing the first electrode; and a first blue emitting material layer including a first compound, a second compound and a third compound and positioned between the first and second electrodes. The first compound is an n-type host, the second compound is a p-type host, and the third compound is a dopant or an auxiliary host.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application claims priority to Korean Patent Application No. 10-2024-0027300 filed in the Republic of Korea on Feb. 26, 2024, which is hereby incorporated by reference in its entirety into the present application.

BACKGROUND

Technical Field

[0002] The present disclosure relates to an organic light emitting diode, and more particularly, to an organic light emitting diode having an advantage in at least one of a driving voltage, an emitting efficiency, a color purity and a lifespan and an organic light emitting device including the organic light emitting diode.

Discussion of the Background Art

[0003] A need for flat panel display devices having small occupied area has increased. Among the flat panel display devices, a technology of an organic light emitting display device, which includes an organic light emitting diode (OLED) and can be called to as an organic electroluminescent device, is rapidly developed.

[0004] The OLED emits light by injecting electrons from a cathode as an electron injection electrode and holes from an anode as a hole injection electrode into an emitting material layer (EML), combining the electrons with the holes, generating an exciton, and transforming the exciton from an excited state to a ground state.

[0005] However, the related art OLED can have a limitation in an emitting performance, e.g., a driving voltage, an emitting efficiency, a color purity and a lifespan. In particular, a blue OLED has a big limitation in the emitting performance.

SUMMARY OF THE DISCLOSURE

[0006] Accordingly, embodiments of the present disclosure are directed to an OLED and an organic light emitting device that substantially obviate one or more of the problems associated with the limitations and disadvantages of the related art.

[0007] An aspect of the present disclosure is to provide an OLED and an organic light emitting device having an advantage in at least one of a driving voltage, an emitting efficiency, a color purity and a lifespan.

[0008] Additional features and aspects will be set forth in the description that follows, and in part will be apparent from the description, or can be learned by practice of the present disclosure concepts provided herein. Other features and aspects of the present disclosure concepts can be realized and attained by the structure particularly pointed out in the written description, or derivable therefrom, and the claims hereof as well as the appended drawings.

[0009] To achieve these and other advantages in accordance with the purpose of the embodiments of the present disclosure, as described herein, an aspect of the present disclosure is an organic light

emitting diode comprising a first electrode; a second electrode facing the first electrode; and a first blue emitting material layer including a first compound, a second compound and a third compound and positioned between the first and second electrodes, wherein the first compound is represented by Formula 1:
##STR00001## [0010] wherein in Formula 1, at least one of R.sub.1 to R.sub.11 is represented by Formula 1-1, at least another one of R.sub.1 to R.sub.11 is represented by Formula 1-2, each of the rest of R.sub.1 to R.sub.11 is independently selected from the group consisting of hydrogen, deuterium, halogen, cyano, a substituted or unsubstituted C1 to C20 alkyl group, a substituted or unsubstituted C3 to C30 cycloalkyl group, a substituted or unsubstituted C1 to C20 alkoxy group, a substituted or unsubstituted C1 to C20 alkylamino group, a substituted or unsubstituted C6 to C30 arylamino group, a substituted or unsubstituted C1 to C20 alkylsilyl group, a substituted or unsubstituted C6 to C30 arylsilyl group, a substituted or unsubstituted C6 to C30 aryl group and a substituted or unsubstituted C3 to C30 heteroaryl group, or optionally, adjacent two of the rest of R.sub.1 to R.sub.11 are combined to form a ring, each of Y.sub.1 and Y.sub.2 is independently selected from O or NR.sub.12, R.sub.12 is selected from the group consisting of a substituted or unsubstituted C1 to C20 alkyl group, a substituted or unsubstituted C3 to C30 cycloalkyl group, a substituted or unsubstituted C1 to C20 alkoxy group, a substituted or unsubstituted C1 to C20 alkylamino group, a substituted or unsubstituted C6 to C30 arylamino group, a substituted or unsubstituted C1 to C20 alkylsilyl group, a substituted or unsubstituted C6 to C30 arylsilyl group, a substituted or unsubstituted C6 to C30 aryl group and a substituted or unsubstituted C3 to C30 heteroaryl group,
##STR00002## [0011] wherein in Formula 1-1, L is selected from a substituted or unsubstituted C6 to C30 arylene group, and each of Ar.sub.1, Ar.sub.2 and Ar.sub.3 is independently selected from a substituted or unsubstituted C6 to C30 aryl group, [0012] wherein in Formula 1-2, each of a1 and a4 is independently an integer of 0 to 4, each of a2 and a3 is independently an integer of 0 to 2, when a1 is 2 or more, two or more R.sub.21 are same or different, when a2 is 2, two R.sub.22 are same or different, when a3 is 2, two R.sub.23 are same or different, when a4 is 2 or more, two or more R.sub.24 are same or different, each of R.sub.21, R.sub.22, R.sub.23 and R.sub.24 is independently selected from the group consisting of deuterium, halogen, cyano, a substituted or unsubstituted C1 to C20 alkyl group, a substituted or unsubstituted C3 to C30 cycloalkyl group, a substituted or unsubstituted C1 to C20 alkoxy group, a substituted or unsubstituted C1 to C20 alkylamino group, a substituted or unsubstituted C6 to C30 arylamino group, a substituted or unsubstituted C1 to C20 alkylsilyl group, a substituted or unsubstituted C6 to C30 arylsilyl group, a substituted or unsubstituted C6 to C30 aryl group and a substituted or unsubstituted C3 to C30 heteroaryl group, each of n1 and n2 is independently 0 or 1, one of X.sub.1 and X.sub.2 is a single bond, the other one of X.sub.1 and X.sub.2 is selected from O, S, Se, NR.sub.25 and C(R.sub.25).sub.2, one of X.sub.3 and X.sub.4 is a single bond, the other one of X.sub.3 and X.sub.4 is selected from O, S, Se, NR.sub.26 and C(R.sub.26).sub.2, and each of R.sub.25 and R.sub.26 is independently selected from the group consisting of hydrogen, deuterium, halogen, cyano, a substituted or unsubstituted C1 to C20 alkyl group, a substituted or unsubstituted C3 to C30 cycloalkyl group, a substituted or unsubstituted C1 to C20 alkoxy group, a substituted or unsubstituted C1 to C20 alkylamino group, a substituted or unsubstituted C6 to C30 arylamino group, a substituted or unsubstituted C1 to C20 alkylsilyl group, a substituted or unsubstituted C6 to C30 arylsilyl group, a substituted or unsubstituted C6 to C30 aryl group and a substituted or unsubstituted C3 to C30 heteroaryl group, wherein the second compound is represented by Formula 3:
##STR00003## [0013] wherein in Formula 3, each of b1, b2, b3 and b4 is independently an integer of 0 to 4, when b1 is 2 or more, two or more R.sub.41 are same or different, when b2 is 2 or more, two or more R.sub.42 are same or different, when b3 is 2 or more, two or more R.sub.43 are same or different, when b4 is 2 or more, two or more R.sub.44 are same or different, and each of

R.sub.41 to R.sub.44 is independently selected from the group consisting of deuterium, halogen, cyano, a substituted or unsubstituted C1 to C20 alkyl group, a substituted or unsubstituted C1 to C20 alkoxy group, a substituted or unsubstituted C1 to C20 alkylamino group, a substituted or unsubstituted C6 to C30 arylamino group, a substituted or unsubstituted C1 to C20 alkylsilyl group, a substituted or unsubstituted C6 to C30 arylsilyl group, a substituted or unsubstituted C6 to C30 aryl group and a substituted or unsubstituted C3 to C30 heteroaryl group, and wherein the third compound is represented by Formula 5:
##STR00004## [0014] wherein in Formula 5, each of d1, d2 and d3 is independently an integer of 0 to 4, d4 is an integer of 0 to 3, d5 is an integer of 0 to 2, when d1 is 2 or more, two or more R.sub.51 are same or different, when d2 is 2 or more, two or more R.sub.52 are same or different, when d3 is 2 or more, two or more R.sub.53 are same or different, when d4 is 2 or more, two or more R.sub.54 are same or different, when d5 is 2, two R.sub.55 are same or different, each of R.sub.51 to R.sub.55 is independently selected from the group consisting of deuterium, halogen, cyano, a substituted or unsubstituted C1 to C20 alkyl group, a substituted or unsubstituted C3 to C30 cycloalkyl group, a substituted or unsubstituted C1 to C20 alkylsilyl group, a substituted or unsubstituted C1 to C20 alkylamino group, a substituted or unsubstituted C6 to C30 arylamino group, a substituted or unsubstituted C6 to C30 arylsilyl group, a substituted or unsubstituted C6 to C30 aryl group and a substituted or unsubstituted C3 to C30 heteroaryl group, and R.sub.56 is selected from the group consisting of hydrogen, deuterium, halogen, cyano, a substituted or unsubstituted C1 to C20 alkyl group, a substituted or unsubstituted C3 to C30 cycloalkyl group, a substituted or unsubstituted C1 to C20 alkylsilyl group, a substituted or unsubstituted C1 to C20 alkylamino group, a substituted or unsubstituted C6 to C30 arylamino group, a substituted or unsubstituted C6 to C30 arylsilyl group, a substituted or unsubstituted C6 to C30 aryl group and a substituted or unsubstituted C3 to C30 heteroaryl group.
[0015] Another aspect of the present disclosure is an organic light emitting device comprising a substrate; the above organic light emitting diode disposed on the substrate; and an encapsulation layer covering the organic light emitting diode.
[0016] It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory and are intended to provide further explanation of the inventive concepts.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0017] The accompanying drawings, which are included to provide a further understanding of the present disclosure and are incorporated in and constitute a part of this application, illustrate embodiments of the present disclosure and together with the description serve to explain principles of the present disclosure.
[0018] FIG. **1** is a schematic circuit diagram of an organic light emitting display device according to one or more embodiments of the present disclosure.
[0019] FIG. **2** is a schematic cross-sectional view of an organic light emitting display device according to a first embodiment of the present disclosure.
[0020] FIG. **3** is a schematic cross-sectional view of an OLED according to a second embodiment of the present disclosure.
[0021] FIG. **4** is a schematic energy band diagram in an emitting material layer (EML) of an OLED according to the second embodiment of the present disclosure.
[0022] FIG. **5** is a schematic cross-sectional view of an OLED according to a third embodiment of the present disclosure.
[0023] FIG. **6** is a schematic energy band diagram in an EML of an OLED according to the third

embodiment of the present disclosure.
[0024] FIG. **7** is a schematic cross-sectional view of an OLED according to a fourth embodiment of the present disclosure.
[0025] FIG. **8** is a schematic cross-sectional view of an OLED according to a fifth embodiment of the present disclosure.
[0026] FIG. **9** is a schematic cross-sectional view of an OLED according to a sixth embodiment of the present disclosure.
[0027] FIG. **10** is a schematic cross-sectional view of an organic light emitting display device according to a seventh embodiment of the present disclosure.
[0028] FIG. **11** is a schematic cross-sectional view of an organic light emitting display device according to an eighth embodiment of the present disclosure.
[0029] FIG. **12** is a schematic cross-sectional view of an OLED according to a ninth embodiment of the present disclosure.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0030] Reference will now be made in detail to aspects of the present disclosure, examples of which can be illustrated in the accompanying drawings. In the following description, when a detailed description of well-known functions or configurations related to this document is determined to unnecessarily cloud a gist of the inventive concept, the detailed description thereof will be omitted. The progression of processing steps and/or operations described is an example; however, the sequence of steps and/or operations is not limited to that set forth herein and can be changed as is known in the art, with the exception of steps and/or operations necessarily occurring in a particular order. Like reference numerals designate like elements throughout the specification. Names of the respective elements used in the following explanations are selected only for convenience of writing the specification and can be thus different from those used in actual products. Further, the term “can” fully encompasses all the meanings and coverages of the term “may.”
[0031] Advantages and features of the present disclosure and methods of achieving them will be apparent with reference to the aspects described below in detail with the accompanying drawings. However, the present disclosure is not limited to the aspects disclosed below, but can be realized in a variety of different forms, and only these aspects allow the disclosure of the present disclosure to be complete. The present disclosure is provided to fully inform the scope of the disclosure to the skilled in the art of the present disclosure.
[0032] The shapes, sizes, proportions, angles, numbers, and the like disclosed in the drawings for explaining the aspects of the present disclosure are illustrative, and the present disclosure is not limited to the illustrated matters. The same reference numerals refer to the same elements throughout the specification. In addition, in describing the present disclosure, if it is determined that a detailed description of the related known technology unnecessarily obscure the subject matter of the present disclosure, the detailed description thereof can be omitted. When ‘including’, ‘having’, ‘consisting’, and the like are used in this specification, other parts can be added unless ‘only’ is used. When a component is expressed in the singular, cases including the plural are included unless specific statement is described.
[0033] In construing an element, the element is construed as including an error or tolerance range although there is no explicit description of such an error or tolerance range.
[0034] In describing a position relationship, for example, when a position relation between two parts is described as, for example, “on,” “over,” “under,” and “next,” one or more other parts can be disposed between the two parts unless a more limiting term, such as “just” or “direct(ly)” is used.
[0035] In describing a time relationship, for example, when the temporal order is described as, for example, “after,” “subsequent,” “next,” and “before,” a case that is not continuous can be included unless a more limiting term, such as “just,” “immediate(ly),” or “direct(ly)” is used.
[0036] It will be understood that, although the terms “first,” “second,” etc. can be used herein to

describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first element could be termed a second element, and, similarly, a second element could be termed a first element, without departing from the scope of the present disclosure.
[0037] Features of various aspects of the present disclosure can be partially or overall coupled to or combined with each other, and can be variously inter-operated with each other and driven technically as those skilled in the art can sufficiently understand. The aspects of the present disclosure can be carried out independently from each other, or can be carried out together in co-dependent relationship.
[0038] A photoluminescence (PL) spectrum can be measured using an organic solvent, e.g., toluene, at room temperature, i.e., 25° C. For example, after a thin film having a thickness of 30 nm is formed using a solution, in which a compound is dissolved in an organic solvent, e.g., toluene, with about 1*10.sup.−5M, a PL spectrum can be measured using a PL detection and a fluorescent spectrometer, e.g., FS-5 fluorescent spectrometer (Edinburgh Instruments).
[0039] Various methods of determining a highest occupied molecular orbital (HOMO) energy level are known to the skilled person. For example, the HOMO energy level can be determined using a conventional surface analyser such as an AC3 surface analyser made by RKI instruments. The surface analyser can be used to interrogate a single film (neat film) of a compound with a thickness of 50 nm. The LUMO energy level can be calculated as follows:

LUMO energy level (eV)=HOMO energy level (eV)−bandgap energy level (eV).
[0040] The bandgap can be measured using a SCINCO S-3100 spectrophotometer. The HOMO and LUMO values of the compounds of the examples and embodiments disclosed herein can be determined in this way. Namely, the HOMO and LUMO values can be experimentally or empirically determined values of thin films, such as 50 nm films.
[0041] Reference will now be made in detail to some of the examples and preferred embodiments, which are illustrated in the accompanying drawings.
[0042] In an OLED and an organic light emitting device, a blue emitting material layer (EML) includes an n-type host, a p-type host and a dopant (e.g., an emitter). For example, the organic light emitting device can be an organic light emitting display device or an organic lightening device. As an example, an organic light emitting display device, which is a display device including the OLED of the present disclosure, will be mainly described.
[0043] FIG. **1** is a schematic circuit diagram of an organic light emitting display device of the present disclosure.
[0044] As shown in FIG. **1**, an organic light emitting display device includes a gate line GL, a data line DL, a power line PL, a switching thin film transistor TFT Ts, a driving TFT Td, a storage capacitor Cst, and an OLED D. The gate line GL and the data line DL cross each other to define a pixel region P. The pixel region P can include a red pixel region, a green pixel region and a blue pixel region.
[0045] The switching TFT Ts is connected to the gate line GL and the data line DL, and the driving TFT Td and the storage capacitor Cst are connected to the switching TFT Ts and the power line PL. The OLED D is connected to the driving TFT Td.
[0046] In the organic light emitting display device, when the switching TFT Ts is turned on by a gate signal applied through the gate line GL, a data signal from the data line DL is applied to the gate electrode of the driving TFT Td and an electrode of the storage capacitor Cst.
[0047] When the driving TFT Td is turned on by the data signal, an electric current is supplied to the OLED D from the power line PL. As a result, the OLED D emits light. In this case, when the driving TFT Td is turned on, a level of an electric current applied from the power line PL to the OLED D is determined such that the OLED D can produce a gray scale.
[0048] The storage capacitor Cst serves to maintain the voltage of the gate electrode of the driving

TFT Td when the switching TFT Ts is turned off. Accordingly, even if the switching TFT Ts is turned off, a level of an electric current applied from the power line PL to the OLED D is maintained to next frame.
[0049] As a result, the organic light emitting display device displays a desired image.
[0050] FIG. **2** is a schematic cross-sectional view of an organic light emitting display device according to a first embodiment of the present disclosure.
[0051] As shown in FIG. **2**, the organic light emitting display device **100** includes a substrate **110**, a TFT Tr on or over the substrate **110**, a planarization layer **150** covering the TFT Tr and an OLED D on the planarization layer **150** and connected to the TFT Tr. A red pixel region, a green pixel region and a blue pixel region can be defined on the substrate **110**.
[0052] The substrate **110** can be a glass substrate or a flexible substrate. For example, the flexible substrate can be one of a polyimide (PI) substrate, a polyethersulfone (PES) substrate, a polyethylenenaphthalate (PEN) substrate, a polyethylene terephthalate (PET) substrate and a polycarbonate (PC) substrate.
[0053] A buffer layer **122** is formed on the substrate **110**, and the TFT Tr is formed on the buffer layer **122**. The buffer layer **122** can be omitted. For example, the buffer layer **122** can be formed of an inorganic insulating material, e.g., silicon oxide or silicon nitride.
[0054] A semiconductor layer **120** is formed on the buffer layer **122**. The semiconductor layer **120** can include an oxide semiconductor material or polycrystalline silicon.
[0055] When the semiconductor layer **120** includes the oxide semiconductor material, a light-shielding pattern can optionally be formed under the semiconductor layer **120**. The light to the semiconductor layer **120** is shielded or blocked by the light-shielding pattern such that thermal degradation of the semiconductor layer **120** can be prevented. On the other hand, when the semiconductor layer **120** includes polycrystalline silicon, impurities can be doped into both sides of the semiconductor layer **120**.
[0056] A gate insulating layer **124** is formed on the semiconductor layer **120**. The gate insulating layer **124** can be formed of an inorganic insulating material such as silicon oxide or silicon nitride.
[0057] A gate electrode **130**, which is formed of a conductive material, e.g., metal, is formed on the gate insulating layer **124** to correspond to a center of the semiconductor layer **120**. In FIG. **2**, the gate insulating layer **124** is formed on an entire surface of the substrate **110**. Alternatively, the gate insulating layer **124** can be patterned to have the same shape as the gate electrode **130**.
[0058] An interlayer insulating layer **132** is formed on the gate electrode **130** and over an entire surface of the substrate **110**. The interlayer insulating layer **132** can be formed of an inorganic insulating material, e.g., silicon oxide or silicon nitride, or an organic insulating material, e.g., benzocyclobutene or photo-acryl.
[0059] The interlayer insulating layer **132** includes first and second contact holes **134** and **136** exposing both sides of the semiconductor layer **120**. The first and second contact holes **134** and **136** are positioned at both sides of the gate electrode **130** to be spaced apart from the gate electrode **130**.
[0060] The first and second contact holes **134** and **136** are formed through the gate insulating layer **124** and the interlayer insulating layer **132**. Alternatively, when the gate insulating layer **124** is patterned to have the same shape as the gate electrode **130**, the first and second contact holes **134** and **136** is formed only through the interlayer insulating layer **132**.
[0061] A source electrode **144** and a drain electrode **146**, which are formed of a conductive material, e.g., metal, are formed on the interlayer insulating layer **132**.
[0062] The source electrode **144** and the drain electrode **146** are spaced apart from each other with respect to the gate electrode **130** and respectively contact both sides of the semiconductor layer **120** through the first and second contact holes **134** and **136**.
[0063] The semiconductor layer **120**, the gate electrode **130**, the source electrode **144** and the drain electrode **146** constitute the TFT Tr. The TFT Tr serves as a driving element. Namely, the TFT Tr is

the driving TFT Td (of FIG. **1**).

[0064] In the TFT Tr, the gate electrode **130**, the source electrode **144**, and the drain electrode **146** are positioned over the semiconductor layer **120**. Namely, the TFT Tr has a coplanar structure.

[0065] Alternatively, in the TFT Tr, the gate electrode can be positioned under the semiconductor layer, and the source and drain electrodes can be positioned over the semiconductor layer such that the TFT Tr can have an inverted staggered structure. In this instance, the semiconductor layer can include amorphous silicon.

[0066] Optionally, the gate line and the data line cross each other to define the pixel region, and the switching TFT is formed to be connected to the gate and data lines. The switching TFT is connected to the TFT Tr as the driving element. In addition, the power line, which can be formed to be parallel to and spaced apart from one of the gate and data lines, and the storage capacitor for maintaining the voltage of the gate electrode of the TFT Tr in one frame can be further formed.

[0067] A planarization layer **150** is formed on an entire surface of the substrate **110** to cover the source and drain electrodes **144** and **146**. The planarization layer **150** provides a flat top surface and has a drain contact hole **152** exposing the drain electrode **146** of the TFT Tr.

[0068] The OLED D is disposed on the planarization layer **150** and includes a first electrode **210**, which is connected to the drain electrode **146** of the TFT Tr, an organic light emitting layer **220** and a second electrode **230**. The organic light emitting layer **220** and the second electrode **230** are sequentially stacked on the first electrode **210**. The OLED D is positioned in each of the red, green and blue pixel regions and respectively emits the red, green and blue light.

[0069] The first electrode **210** is separately formed in each pixel region. The first electrode **210** can be an anode and can include a transparent conductive oxide material layer, which can be formed of a conductive material, e.g., a transparent conductive oxide (TCO), having a relatively high work function.

[0070] For example, the transparent conductive oxide material layer can be formed of one of indium-tin-oxide (ITO), indium-zinc-oxide (IZO), indium-tin-zinc-oxide (ITZO), tin oxide (SnO), zinc oxide (ZnO), indium-copper-oxide (ICO) and aluminum-zinc-oxide (Al:ZnO, AZO).

[0071] The first electrode **210** can have a single-layered structure of the transparent conductive oxide material layer. Namely, the first electrode **210** can be a transparent electrode.

[0072] Alternatively, the first electrode **210** can further include a reflective layer to have a double-layered structure or a triple-layered structure. Namely, the first electrode **210** can be a reflective electrode.

[0073] For example, the reflective layer can be formed of one of silver (Ag), an alloy of Ag and one of palladium (Pd), copper (Cu), indium (In) and neodymium (Nd), and aluminum-palladium-copper (APC) alloy. For example, the first electrode **210** can have a double-layered structure of Ag/ITO or APC/ITO or a triple-layered structure of ITO/Ag/ITO or ITO/APC/ITO.

[0074] In addition, a bank layer **160** is formed on the planarization layer **150** to cover an edge of the first electrode **210**. Namely, the bank layer **160** is positioned at a boundary of the pixel region and exposes a center of the first electrode **210** in the pixel region.

[0075] The organic light emitting layer **220** including an emitting material layer (EML) is formed on the first electrode **210**. In the OLED D in the blue pixel region, the EML of the organic light emitting layer **220** includes a first compound represented by Formula 1, a second compound represented by Formula 3 and a third compound represented by Formula 5. In addition, the EML can further include a fluorescent compound represented by Formula 7.

[0076] The organic light emitting layer **220** can further include at least one of a hole injection layer (HIL), a hole transporting layer (HTL), an electron blocking layer (EBL), a hole blocking layer (HBL), an electron transporting layer (ETL) and an electron injection layer (EIL) to have a multi-layered structure.

[0077] In an aspect of the present disclosure, the organic light emitting layer **220** of the OLED D in the blue pixel region can include a first blue emitting part including a first blue EML and a second blue emitting part including a second blue EML to have a tandem structure, and at least one of the first and second blue EMLs includes the first compound represented by Formula 1, the second compound represented by Formula 3 and the third compound represented by Formula 5. In addition, at least one of the first and second blue EMLs can further include the fluorescent compound represented by Formula 7. In this case, the organic light emitting layer **220** can further include a charge generation layer (CGL) between the first and second blue emitting parts.

[0078] The second electrode **230** is formed over the substrate **110** where the organic light emitting layer **220** is formed. The second electrode **230** covers an entire surface of the display area and can be formed of a conductive material having a relatively low work function to serve as a cathode. For example, the second electrode **230** can be formed of aluminum (Al), magnesium (Mg), calcium (Ca), silver (Ag) or their alloy, e.g., Mg—Ag alloy (MgAg).

[0079] In a top-emission type OLED D, the first electrode **210** serves as a reflective electrode, and the second electrode **230** has a thin profile to serve as a transparent (or a semi-transparent) electrode. Alternatively, in a bottom-emission type OLED D, the first electrode **210** serves as a transparent electrode, and the second electrode **230** serves as a reflective electrode.

[0080] Optionally, the top-emission type OLED D can further include a capping layer on the second electrode **230**. The emitting efficiency of the OLED D and the organic light emitting display device **100** including the OLED D can be further improved by the capping layer.

[0081] An encapsulation layer (or an encapsulation film) **170** is formed on the second electrode **230** to prevent penetration of moisture into the OLED D. The encapsulation layer **170** includes a first inorganic insulating layer **172**, an organic insulating layer **174** and a second inorganic insulating layer **176** sequentially stacked, but it is not limited thereto.

[0082] In the bottom-emission type organic light emitting display device **100**, a metal encapsulation plate can be disposed over the second electrode **230** or the encapsulation layer **170**. For example, the metal encapsulation plate can be attached to the second electrode **230** or the encapsulation layer **170** using an adhesive layer.

[0083] Optionally, the organic light emitting display device **100** can include a color filter layer corresponding to the red, green and blue pixel regions. In the top-emission type organic light emitting display device **100**, the color filter layer can be disposed over the OLED D. In the bottom-emission type organic light emitting display device **100**, the color filter layer can be disposed between the substrate **110** and the OLED D.

[0084] The color filter layer can include red, green and blue color filters respectively corresponding to the red, green and blue pixel regions. The red color filter can include at least one of a red dye and a red pigment, the green color filter can include at least one of a green dye and a green pigment, and the blue color filter can include at least one of a blue dye and a blue pigment.

[0085] The organic light emitting display device **100** can further include a polarization plate for reducing an ambient light reflection. For example, the polarization plate can be a circular polarization plate. In the bottom-emission type organic light emitting display device **100**, the polarization plate can be disposed under the substrate **110**. In the top-emission type organic light emitting display device **100**, the polarization plate can be disposed on or over the encapsulation layer **170**.

[0086] In addition, the organic light emitting display device **100** can further include a cover window on or over the encapsulation layer **170** or the polarization plate. In this instance, the substrate **110** and the cover window have a flexible property such that a flexible organic light emitting display device can be provided.

[0087] FIG. **3** is a schematic cross-sectional view of an OLED according to a second embodiment of the present disclosure.

[0088] As shown in FIG. **3**, the OLED D**1** includes first and second electrodes **210** and **230**, which face each other, and an organic light emitting layer **220** therebetween. The organic light emitting layer **220** includes an emitting material layer (EML) **240**A, e.g., a blue EML. The OLED D**1** can

further include a capping layer on the second electrode **230** to enhance a light extraction efficiency.
[0089] The organic light emitting display device **100** (of FIG. **2**) can include a red pixel region, a green pixel region and a blue pixel region, and the OLED D**1** can be positioned in the blue pixel region.
[0090] The first electrode **210** can be an anode, and the second electrode **230** can be a cathode. One of the first and second electrodes **210** and **230** can be a reflective electrode, and the other one of the first and second electrodes **210** and **230** can be a transparent (or a semi-transparent) electrode. In the bottom-emission type OLED D**1**, the first electrode **210** can have a single-layered structure of ITO, and the second electrode **230** can be formed of Al. The first electrode **210** can have a thickness of 10 nm to 100 nm, e.g., 40 nm to 60 nm, and the second electrode **230** can have a thickness of 100 nm to 300 nm, e.g., 150 nm to 250 nm.
[0091] The organic light emitting layer **220** can further include at least one of a hole transporting layer (HTL) **260** between the first electrode **210** and the EML **240**A and an electron transporting layer (ETL) **270** between the second electrode **230** and the EML **240**A.
[0092] In addition, the organic light emitting layer **220** can further include at least one of a hole injection layer (HIL) **250** between the first electrode **210** and the HTL **260** and an electron injection layer (EIL) **280** between the second electrode **230** and the ETL **270**.
[0093] Moreover, the organic light emitting layer **220** can further include at least one of an electron blocking layer (EBL) **265** between the HTL **260** and the EML **240**A and a hole blocking layer (HBL) **275** between the EML **240**A and the ETL **270**.
[0094] For example, the HIL **250** can include a hole injection material being at least one compound selected from the group consisting of 4,4′,4″-tris(3-methylphenylamino)triphenylamine (MTDATA), 4,4′,4″-tris(N,N-diphenyl-amino)triphenylamine (NATA), 4,4′,4″-tris(N-(naphthalene-1-yl)-N-phenyl-amino)triphenylamine (1T-NATA), 4,4′,4″-tris(N-(naphthalene-2-yl)-N-phenyl-amino)triphenylamine (2T-NATA), copper phthalocyanine(CuPc), tris(4-carbazoyl-9-yl-phenyl)amine (TCTA), N,N′-diphenyl-N,N′-bis(1-naphthyl)-1,1′-biphenyl-4,4″-diamine (NPB or NPD), 1,4,5,8,9,11-hexaazatriphenylenehexacarbonitrile(dipyrazino[2,3-f:2′3′-h]quinoxaline-2,3,6,7,10,11-hexacarbonitrile) (HAT-CN), 1,3,5-tris[4-(diphenylamino)phenyl]benzene (TDAPB), poly(3,4-ethylenedioxythiphene)polystyrene sulfonate (PEDOT/PSS), and N-(biphenyl-4-yl)-9,9-dimethyl-N-(4-(9-phenyl-9H-carbazol-3-yl)phenyl)-9H-fluoren-2-amine, but it is not limited thereto. In an aspect of the present disclosure, the hole injection material of the HIL **250** can be a compound in Formula 9. For example, the HIL **250** can have a thickness of 1 nm to 20 nm, e.g., 5 nm to 15 nm.
[0095] The HTL **260** can include a hole transporting material being at least one compound selected from the group consisting of N,N′-diphenyl-N,N′-bis(3-methylphenyl)-1,1′-biphenyl-4,4′-diamine (TPD), NPB(NPD), 4,4′-bis(N-carbazolyl)-1,1′-biphenyl (CBP), poly[N,N′-bis(4-butylphenyl)-N,N′-bis(phenyl)-benzidine](poly-TPD), poly[(9,9-dioctylfluorenyl-2,7-diyl)-co-(4,4′-(N-(4-sec-butylphenyl)diphenylamine))](TFB), di-[4-(N,N-di-p-tolyl-amino)-phenyl]cyclohexane (TAPC), 3,5-di(9H-carbazol-9-yl)-N,N-diphenylaniline (DCDPA), N-(biphenyl-4-yl)-9,9-dimethyl-N-(4-(9-phenyl-9H-carbazol-3-yl)phenyl)-9H-fluoren-2-amine, and N-(biphenyl-4-yl)-N-(4-(9-phenyl-9H-carbazol-3-yl)phenyl)biphenyl-4-amine, but it is not limited thereto. In an aspect of the present disclosure, the hole transporting material of the HTL **260** can be a compound in Formula 10. For example, the HTL **260** can have a thickness of 30 nm to 60 nm, e.g., 40 nm to 50 nm.
[0096] The ETL **270** can include an electron transporting material being at least one of an oxadiazole-based compound, a triazole-based compound, a phenanthroline-based compound, a benzoxazole-based compound, a benzothiazole-based compound, a benzimidazole-based compound, and a triazine-based compound. For example, the ETL **270** can include at least one compound selected from the group consisting of tris-(8-hydroxyquinoline) aluminum (Alq.sub.3), 2-biphenyl-4-yl-5-(4-t-butylphenyl)-1,3,4-oxadiazole (PBD), spiro-PBD, lithium quinolate (Liq), 1,3,5-tris(N-phenylbenzimidazol-2-yl)benzene (TPBi), bis(2-methyl-8-quinolinolato-N1,08)-(1,1′-

biphenyl-4-olato)aluminum (BAlq), 4,7-diphenyl-1,10-phenanthroline (Bphen), 2,9-bis(naphthalene-2-yl)4,7-diphenyl-1,10-phenanthroline (NBphen), 2,9-dimethyl-4,7-diphenyl-1,10-phenathroline (BCP), 3-(4-biphenyl)-4-phenyl-5-tert-butylphenyl-1,2,4-triazole (TAZ), 4-(naphthalen-1-yl)-3,5-diphenyl-4H-1,2,4-triazole (NTAZ), 1,3,5-tri(p-pyrid-3-yl-phenyl)benzene (TpPyPB), 2,4,6-tris(3′-(pyridin-3-yl)biphenyl-3-yl)1,3,5-triazine (TmPPPyTz), Poly[9,9-bis(3′-((N,N-dimethyl)-N-ethylammonium)-propyl)-2,7-fluorene]-alt-2,7-(9,9-dioctylfluorene)](PFNBr), tris(phenylquinoxaline) (TPQ), and diphenyl-4-triphenylsilyl-phenylphosphine oxide (TSPO1), but it is not limited thereto. In an aspect of the present disclosure, the electron transporting material of the ETL **270** can be a compound in Formula 13. For example, the ETL **270** can have a thickness of 10 nm to 50 nm, e.g., 20 nm to 40 nm.

[0097] The EIL **280** can include an electron injection material being at least one of an alkali halide compound, such as LiF, CsF, NaF, or BaF.sub.2, and an organo-metallic compound, such as Liq, lithium benzoate, or sodium stearate, but it is not limited thereto. For example, the EIL **280** can have a thickness of 0.1 nm to 10 nm, e.g., 1 nm to 5 nm.

[0098] The EBL **265**, which is positioned between the HTL **260** and the EML **240**A to block the electron transfer from the EML **240**A into the HTL **260**, can include an electron blocking material being at least one compound selected from the group consisting of TCTA, tris[4-(diethylamino)phenyl]amine, N-(biphenyl-4-yl)-9,9-dimethyl-N-(4-(9-phenyl-9H-carbazol-3-yl)phenyl)-9H-fluoren-2-amine, TAPC, MTDATA, 1,3-bis(carbazol-9-yl)benzene (mCP), 3,3′-bis(N-carbazolyl)-1,1′-biphenyl (mCBP), CuPc, N,N′-bis[4-[bis(3-methylphenyl)amino]phenyl]-N,N′-diphenyl-[1,1′-biphenyl]-4,4′-diamine (DNTPD), TDAPB, DCDPA, and 2,8-bis(9-phenyl-9H-carbazol-3-yl)dibenzo[b,d]thiophene, but it is not limited thereto. In an aspect of the present disclosure, the electron blocking material of the EBL **265** can be a compound in Formula 11. For example, the EBL **265** can have a thickness of 5 nm to 20 nm, e.g., 5 nm to 15 nm.

[0099] The HBL **275**, which is positioned between the EML **240**A and the ETL **270** to block the hole transfer from the EML **240**A into the ETL **270**, can include the above material of the ETL **270**. For example, the material of the HBL **275** has a HOMO energy level being lower than a material of the EML **240**A and can be at least one compound selected from the group consisting of BCP, BAlq, Alq.sub.3, PBD, spiro-PBD, Liq, bis-4,6-(3,5-di-3-pyridylphenyl)-2-methylpyrimidine (B3PYMPM), bis[2-(diphenylphosphino)phenyl]ether oxide (DPEPO), (9-(6-9H-carbazol-9-yl)pyridine-3-yl)-9H-3,9′-bicarbazole, and TSPO1, but it is not limited thereto. In an aspect of the present disclosure, the hole blocking material of the HBL **275** can be a compound in Formula 12. For example, the HBL **275** can have a thickness of 5 nm to 20 nm, e.g., 5 nm to 15 nm.

[0100] The EML **240**A can have a thickness of 10 nm to 100 nm, e.g., 10 nm to 50 nm, preferably 20 nm to 40 nm. The EML **240**A can include a first compound **242***a*, a second compound **244***a* and a third compound **246***a*. The first compound **242***a* can be an n-type host, the second compound **244***a* can be a p-type host, and the third compound **246***a* can be a dopant (e.g., an emitter).

[0101] The first compound **242***a* is represented by Formula 1.

##STR00005##

[0102] In Formula 1, [0103] at least one of R.sub.1 to R.sub.11 is represented by Formula 1-1, at least another one of R.sub.1 to R.sub.11 is represented by Formula 1-2, [0104] each of the rest of R.sub.1 to R.sub.11 is independently selected from the group consisting of hydrogen, deuterium, halogen, cyano, a substituted or unsubstituted C1 to C20 alkyl group, a substituted or unsubstituted C3 to C30 cycloalkyl group, a substituted or unsubstituted C1 to C20 alkoxy group, a substituted or unsubstituted C1 to C20 alkylamino group, a substituted or unsubstituted C6 to C30 arylamino group, a substituted or unsubstituted C1 to C20 alkylsilyl group, a substituted or unsubstituted C6 to C30 arylsilyl group, a substituted or unsubstituted C6 to C30 aryl group and a substituted or unsubstituted C3 to C30 heteroaryl group, or optionally, adjacent two of the rest of R.sub.1 to R.sub.11 are combined to form a ring, [0105] each of Y.sub.1 and Y.sub.2 is independently selected from O or NR.sub.12, [0106] R.sub.12 is selected from the group consisting of a substituted or

unsubstituted C1 to C20 alkyl group, a substituted or unsubstituted C3 to C30 cycloalkyl group, a substituted or unsubstituted C1 to C20 alkoxy group, a substituted or unsubstituted C1 to C20 alkylamino group, a substituted or unsubstituted C6 to C30 arylamino group, a substituted or unsubstituted C1 to C20 alkylsilyl group, a substituted or unsubstituted C6 to C30 arylsilyl group, a substituted or unsubstituted C6 to C30 aryl group and a substituted or unsubstituted C3 to C30 heteroaryl group,
##STR00006## [0107] in Formula 1-1, [0108] L is selected from a substituted or unsubstituted C6 to C30 arylene group, [0109] each of Ar.sub.1, Ar.sub.2 and Ar.sub.3 is independently selected from a substituted or unsubstituted C6 to C30 aryl group, [0110] in Formula 1-2, [0111] each of a1 and a4 is independently an integer of 0 to 4, each of a2 and a3 is independently an integer of 0 to 2, [0112] when a1 is 2 or more, two or more R.sub.21 are same or different, when a2 is 2, two R.sub.22 are same or different, when a3 is 2, two R.sub.23 are same or different, when a4 is 2 or more, two or more R.sub.24 are same or different, [0113] each of R.sub.21, R.sub.22, R.sub.23 and R.sub.24 is independently selected from the group consisting of deuterium, halogen, cyano, a substituted or unsubstituted C1 to C20 alkyl group, a substituted or unsubstituted C3 to C30 cycloalkyl group, a substituted or unsubstituted C1 to C20 alkoxy group, a substituted or unsubstituted C1 to C20 alkylamino group, a substituted or unsubstituted C6 to C30 arylamino group, a substituted or unsubstituted C1 to C20 alkylsilyl group, a substituted or unsubstituted C6 to C30 arylsilyl group, a substituted or unsubstituted C6 to C30 aryl group and a substituted or unsubstituted C3 to C30 heteroaryl group, [0114] each of n1 and n2 is independently 0 or 1, [0115] one of X.sub.1 and X.sub.2 is a single bond, the other one of X.sub.1 and X.sub.2 is selected from O, S, Se, NR.sub.25 and C(R.sub.25).sub.2, one of X.sub.3 and X.sub.4 is a single bond, the other one of X.sub.3 and X.sub.4 is selected from O, S, Se, NR.sub.26 and C(R.sub.26).sub.2, and [0116] each of R.sub.25 and R.sub.26 is independently selected from the group consisting of hydrogen, deuterium, halogen, cyano, a substituted or unsubstituted C1 to C20 alkyl group, a substituted or unsubstituted C3 to C30 cycloalkyl group, a substituted or unsubstituted C1 to C20 alkoxy group, a substituted or unsubstituted C1 to C20 alkylamino group, a substituted or unsubstituted C6 to C30 arylamino group, a substituted or unsubstituted C1 to C20 alkylsilyl group, a substituted or unsubstituted C6 to C30 arylsilyl group, a substituted or unsubstituted C6 to C30 aryl group and a substituted or unsubstituted C3 to C30 heteroaryl group.
[0117] In each of Formulas 1-1 and 1-2, the mark “*” denotes a bonding site.
[0118] In the present disclosure, without specific definition, when an alkyl group, an alkoxy group, a cycloalkyl group, an alkylamino group, an alkylsilyl group, an alkenyl group, an alkynyl group, an arylamino group, an arylsilyl group, an aryloxy group, an aryl group and a heteroaryl group are substituted, a substituent can be deuterium, halogen, cyano, a cycloalkyl group, an arylamino group, an alkylamino group, an alkyl group unsubstituted or substituted with at least one of deuterium and halogen, an alkoxy group unsubstituted or substituted with at least one of deuterium and halogen, an alkylsilyl group unsubstituted or substituted with at least one of deuterium and halogen, an alkoxysilyl group unsubstituted or substituted with at least one of deuterium and halogen, an arylsilyl group unsubstituted or substituted with at least one of deuterium and halogen, an aryl group unsubstituted or substituted with at least one of deuterium and halogen, and a heteroaryl group unsubstituted or substituted with at least one of deuterium and halogen.
[0119] In the present disclosure, without specific definition, an alkyl group can include a linear alkyl group and a branched alkyl group, and can include a C1 to C20 alkyl group and a C1 to C10 alkyl group. The alkyl group can be selected from the group consisting of methyl, ethyl, propyl, butyl and tert-butyl.
[0120] In the present disclosure, without specific definition, a C3 to C30 cycloalkyl group can include a C3 to C20 cycloalkyl group and a C3 to C10 cycloalkyl group, and can be selected from the group consisting of cyclopropyl, cyclobutyl, cyclohexyl and adamantanyl.
[0121] In the present disclosure, without specific definition, a C6 to C30 arylsilyl group can be

triphenylsilyl.

[0122] In the present disclosure, without specific definition, a ring, which is formed by adjacent two substituents, can be one of a substituted or unsubstituted C3 to C30 alicyclic ring, a substituted or unsubstituted C6 to C30 aromatic ring such as a phenyl ring, and a substituted or unsubstituted C3 to C30 heteroaromatic ring.

[0123] In the present disclosure, without specific definition, a C6 to C30 aryl group can include a C6 to C20 aryl group and can be selected from the group consisting of phenyl, biphenyl, terphenyl, naphthyl, anthracenyl, pentalenyl, indenyl, indenoindenyl, heptalenyl, biphenylenyl, indacenyl, phenanthrenyl, benzophenanthrenyl, dibenzophenanthrenyl, azulenyl, pyrenyl, fluoranthenyl, triphenylenyl, chrysenyl, tetraphenyl, tetracenyl, picenyl, pentaphenyl, pentacenyl, fluorenyl, indenofluorenyl and spiro-fluorenyl.

[0124] In the present disclosure, without specific definition, the C6 to C30 arylene group can include a C6 to C20 arylene group and can be selected from the group consisting of phenylene, biphenylene, terphenylene, naphthylene, anthracenylene, pentalenylene, indenylene, indenoindenylene, heptalenylene, biphenylenylene, indacenylene, phenanthrenylene, benzophenanthrenylene, dibenzophenanthrenylene, azulenylene, pyrenylene, fluoranthenylene, triphenylenylene, chrysenylene, tetraphenylene, tetracenylene, picenylene, pentaphenylene, pentacenylene, fluorenylene, indenofluorenylene, and spiro-fluorenylene.

[0125] In the present disclosure, without specific definition, a C3 to C30 heteroaryl group can include a C3 to C20 heteroaryl group and can be selected from the group consisting of pyrrolyl, pyridinyl, pyrimidinyl, pyrazinyl, pyridazinyl, triazinyl, tetrazinyl, imidazolyl, pyrazolyl, indolyl, isoindolyl, indazolyl, indolizinyl, pyrrolizinyl, carbazolyl, benzocarbazolyl, dibenzocarbazolyl, indolocarbazolyl, indenocarbazolyl, benzofurocarbazolyl, benzothienocarbazolyl, quinolinyl, isoquinolinyl, phthalazinyl, quinoxalinyl, cinnolinyl, quinazolinyl, quinozolinyl, purinyl, benzoquinolinyl, benzoisoquinolinyl, benzoquinazolinyl, benzoquinoxalinyl, acridinyl, phenanthrolinyl, perimidinyl, phenanthridinyl, pteridinyl, naphtharidinyl, furanyl, oxazinyl, oxazolyl, oxadiazolyl, triazolyl, dioxynyl, benzofuranyl, dibenzofuranyl, thiopyranyl, xanthenyl, chromanyl, isochromanyl, thioazinyl, thiophenyl, benzothiophenyl, dibenzothiophenyl, difuropyrazinyl, benzofurodibenzofuranyl, benzothienobenzothiophenyl, benzothienodibenzothiophenyl, benzothienobenzofuranyl, and benzothienodibenzofuranyl.

[0126] In the present disclosure, without specific definition, the C3 to C30 heteroarylene group can include a C3 to C20 heteroarylene group and can be selected from the group consisting of pyrrolylene, pyridinylene, pyrimidinylene, pyrazinylene, pyridazinylene, triazinylene, tetrazinylene, imidazolylene, pyrazolylene, indolylene, isoindolylene, indazolylene, indolizinylene, pyrrolizinylene, carbazolylene, benzocarbazolylene, dibenzocarbazolylene, indolocarbazolylene, indenocarbazolylene, benzofurocarbazolylene, benzothienocarbazolylene, quinolinylene, isoquinolinylene, phthalazinylene, quinoxalinylene, cinnolinylene, quinazolinylene, quinozolinylene, purinylene, benzoquinolinylene, benzoisoquinolinylene, benzoquinazolinylene, benzoquinoxalinylene, acridinylene, phenanthrolinylene, perimidinylene, phenanthridinylene, pteridinylene, cinnolinylene, naphtharidinylene, furanylene, oxazinylene, oxazolylene, oxadiazolylene, triazolylene, dioxynylene, benzofuranylene, dibenzofuranylene, thiopyranylene, xanthenylene, chromanylene, isochromanylene, thioazinylene, thiophenylene, benzothiophenylene, dibenzothiophenylene, difuropyrazinylene, benzofurodibenzofuranylene, benzothienobenzothiophenylene, benzothienodibenzothiophenylene, benzothienobenzofuranylene, and benzothienodibenzofuranylene.

[0127] In an aspect of the present disclosure, one of R.sub.1 and R.sub.2 can have a structure of Formula 1-1, and the other one of R.sub.1 and R.sub.2 can have a structure of Formula 1-2.

[0128] In an aspect of the present disclosure, R.sub.1 can have a structure of Formula 1-1, and R.sub.2 can have a structure of Formula 1-2.

[0129] In an aspect of the present disclosure, one of R.sub.2 and R.sub.3 can have a structure of

Formula 1-1, and the other one of R.sub.2 and R.sub.3 can have a structure of Formula 1-2.
[0130] In an aspect of the present disclosure, R.sub.3 can have a structure of Formula 1-1, and R.sub.2 can have a structure of Formula 1-2. In an aspect of the present disclosure, one of R.sub.2 and R.sub.4 can have a structure of Formula 1-1, and the other one of R.sub.2 and R.sub.4 can have a structure of Formula 1-2. In an aspect of the present disclosure, R.sub.4 can have a structure of Formula 1-1, and R.sub.2 can have a structure of Formula 1-2.
[0131] In an aspect of the present disclosure, one of Y.sub.1 and Y.sub.2 can be O, and in particular, both Y.sub.1 and Y.sub.2 can be O.
[0132] In an aspect of the present disclosure, one of R.sub.1 to R.sub.4 can have a structure of Formula 1-2, and one of R.sub.5 to R.sub.8 can have a structure of Formula 1-1.
[0133] In an aspect of the present disclosure, one of R.sub.1 to R.sub.4 can have a structure of Formula 1-2, and one of R.sub.9 to R.sub.11 can have a structure of Formula 1-1.
[0134] In an aspect of the present disclosure, one of R.sub.1 to R.sub.11 can have a structure of Formula 1-1, and two of R.sub.1 to R.sub.11 can have a structure of Formula 1-2.
[0135] In an aspect of the present disclosure, one of R.sub.1 to R.sub.4 can have a structure of Formula 1-1, and the other two of R.sub.1 to R.sub.4 can have a structure of Formula 1-2.
[0136] In an aspect of the present disclosure, one of R.sub.1 to R.sub.4 can have a structure of Formula 1-1, and the other one of R.sub.1 to R.sub.4 and one of R.sub.9 to R.sub.11 can have a structure of Formula 1-2.
[0137] In an aspect of the present disclosure, one of R.sub.1 to R.sub.4 can have a structure of Formula 1-1, and the other one of R.sub.1 to R.sub.4 and one of R.sub.5 to R.sub.8 can have a structure of Formula 1-2.
[0138] In Formula 1-1, L can be a substituted or unsubstituted phenylene group, and each of Ar.sub.1, Ar.sub.2 and Ar.sub.3 can independently be a substituted or unsubstituted phenyl group. Namely, the structure of Formula 1-1 can be a structure of tetraphenylsilane. For example, Formula 1-1 can be represented by Formula 1-1a.
##STR00007##
[0139] In Formula 1-1a, [0140] a6 is an integer of 0 to 4, each of a7, a8 and a9 is independently an integer of 0 to 5, [0141] when a6 is 2 or more, two or more R.sub.31 are same or different, when a7 is 2 or more, two or more R.sub.32 are same or different, when a8 is 2 or more, two or more R.sub.33 are same or different, when a9 is 2 or more, two or more R.sub.34 are same or different, and [0142] each of R.sub.31, R.sub.32, R.sub.33 and R.sub.34 is independently selected from the group consisting of deuterium, halogen, cyano, a substituted or unsubstituted C1 to C20 alkyl group, a substituted or unsubstituted C3 to C30 cycloalkyl group, a substituted or unsubstituted C1 to C20 alkoxy group, a substituted or unsubstituted C1 to C20 alkylamino group, a substituted or unsubstituted C6 to C30 arylamino group, a substituted or unsubstituted C1 to C20 alkylsilyl group, a substituted or unsubstituted C6 to C30 arylsilyl group, a substituted or unsubstituted C6 to C30 aryl group and a substituted or unsubstituted C3 to C30 heteroaryl group.
[0143] In Formula 1-1a, the mark “*” denotes a bonding site.
[0144] In Formula 1-2, each of n1 and n2 can be 0.
[0145] In Formula 1-2, n1 can be 0, n2 can be 1, one of X.sub.3 and X.sub.4 can be a single bond, and the other one of X.sub.3 and X.sub.4 can be NR.sub.26. In this case, R.sub.26 can be a substituted or unsubstituted C6 to C30 aryl group, e.g., phenyl. For example, the structure of Formula 1-2 can be represented by one of Formulas 1-2a to 1-2g.
##STR00008## ##STR00009##
[0146] In Formula 1-2a, the definitions of R.sub.22, R.sub.23, a2 and a3 are same as those in Formula 1-2, [0147] in each of Formulas 1-2b to 1-2g, the definitions of R.sub.22 to R.sub.24, and a2 to a4 are same as those in Formula 1-2, [0148] a5 is an integer of 0 to 5, [0149] when a5 is 2 or more, two or more R.sub.27 are same or different, and [0150] R.sub.27 is selected from the group consisting of deuterium, halogen, cyano, a substituted or unsubstituted C1 to C20 alkyl group, a

substituted or unsubstituted C3 to C30 cycloalkyl group, a substituted or unsubstituted C1 to C20 alkoxy group, a substituted or unsubstituted C1 to C20 alkylamino group, a substituted or unsubstituted C6 to C30 arylamino group, a substituted or unsubstituted C1 to C20 alkylsilyl group, a substituted or unsubstituted C6 to C30 arylsilyl group, a substituted or unsubstituted C6 to C30 aryl group and a substituted or unsubstituted C3 to C30 heteroaryl group.
[0151] In each of Formulas 1-2a to 1-2g, the mark "*" denotes a bonding site.
[0152] In Formula 1-2a, each of a2 and a3 can be independently 0 or 1. In an aspect of the present disclosure, each of R.sub.22 and R.sub.23 can be independently carbazolyl.
[0153] In each of Formulas 1-2b to 1-2g, each of a2 to a5 can be 0.
[0154] For example, the first compound **242***a* can be one of the compounds in Formula 2.
##STR00010## ##STR00011## ##STR00012## ##STR00013## ##STR00014## ##STR00015## ##STR00016## ##STR00017## ##STR00018## ##STR00019## ##STR00020## ##STR00021## ##STR00022## ##STR00023## ##STR00024## ##STR00025## ##STR00026## ##STR00027## ##STR00028##
##STR00029## ##STR00030##
[0155] The first compound **242***a* can have a lowest unoccupied molecular orbital (LUMO) energy level being in a range −2.60 eV to −2.50 eV and a highest occupied molecular orbital (HOMO) energy level being in a range −5.65 eV to −5.55 eV. In the first compound **242***a*, a difference between a singlet energy level and a triplet energy level can be 0.3 eV or less, e.g., 0.15 eV to 0.25 eV.
[0156] The second compound **244***a* is represented by Formula 3.
##STR00031##
[0157] In Formula 3, each of b1, b2, b3 and b4 is independently an integer of 0 to 4, [0158] when b1 is 2 or more, two or more R.sub.41 are same or different, when b2 is 2 or more, two or more R.sub.42 are same or different, when b3 is 2 or more, two or more R.sub.43 are same or different, when b4 is 2 or more, two or more R.sub.44 are same or different, and [0159] each of R.sub.41 to R.sub.44 is independently selected from the group consisting of deuterium, halogen, cyano, a substituted or unsubstituted C1 to C20 alkyl group, a substituted or unsubstituted C1 to C20 alkoxy group, a substituted or unsubstituted C1 to C20 alkylamino group, a substituted or unsubstituted C6 to C30 arylamino group, a substituted or unsubstituted C1 to C20 alkylsilyl group, a substituted or unsubstituted C6 to C30 arylsilyl group, a substituted or unsubstituted C6 to C30 aryl group and a substituted or unsubstituted C3 to C30 heteroaryl group.
[0160] In an aspect of the present disclosure, each of R.sub.41 to R.sub.44 can be independently selected from the group consisting of a substituted or unsubstituted C1 to C20 alkyl group, e.g., methyl, a substituted or unsubstituted C6 to C30 aryl group, e.g., phenyl, and a substituted or unsubstituted C3 to C30 heteroaryl group, e.g., carbazolyl.
[0161] In an aspect of the present disclosure, each of b1 to b4 can be independently 0 or 1.
[0162] In Formula 3, a position of a carbazole moiety can be specified. For example, Formula 3 can be represented by Formula 3a or Formula 3b.
##STR00032##
[0163] In each of Formulas 3a and 3b, the definitions of R.sub.41, R.sub.42, R.sub.43, R.sub.44, b1, b2, b3 and b4 are same as those in Formula 3.
[0164] For example, the second compound **244***a* can be one of the compounds in Formula 4.
##STR00033## ##STR00034## ##STR00035##
[0165] The second compound **244***a* can have a LUMO energy level being in a range −2.45 eV to −2.40 eV and a HOMO energy level being in a range −5.90 eV to −5.75 eV.
[0166] The third compound **246***a* is represented by Formula 5.
##STR00036##
[0167] In Formula 5, [0168] each of d1, d2 and d3 is independently an integer of 0 to 4, d4 is an integer of 0 to 3, d5 is an integer of 0 to 2, [0169] when d1 is 2 or more, two or more R.sub.51 are

same or different, when d2 is 2 or more, two or more R.sub.52 are same or different, when d3 is 2 or more, two or more R.sub.53 are same or different, when d4 is 2 or more, two or more R.sub.54 are same or different, when d5 is 2, two R.sub.55 are same or different, [0170] each of R.sub.51 to R.sub.55 is independently selected from the group consisting of deuterium, halogen, cyano, a substituted or unsubstituted C1 to C20 alkyl group, a substituted or unsubstituted C3 to C30 cycloalkyl group, a substituted or unsubstituted C1 to C20 alkylsilyl group, a substituted or unsubstituted C1 to C20 alkylamino group, a substituted or unsubstituted C6 to C30 arylamino group, a substituted or unsubstituted C6 to C30 arylsilyl group, a substituted or unsubstituted C6 to C30 aryl group and a substituted or unsubstituted C3 to C30 heteroaryl group, and [0171] R.sub.56 is selected from the group consisting of hydrogen, deuterium, halogen, cyano, a substituted or unsubstituted C1 to C20 alkyl group, a substituted or unsubstituted C3 to C30 cycloalkyl group, a substituted or unsubstituted C1 to C20 alkylsilyl group, a substituted or unsubstituted C1 to C20 alkylamino group, a substituted or unsubstituted C6 to C30 arylamino group, a substituted or unsubstituted C6 to C30 arylsilyl group, a substituted or unsubstituted C6 to C30 aryl group and a substituted or unsubstituted C3 to C30 heteroaryl group.

[0172] In an aspect of the present disclosure, each of R.sub.51 to R.sub.56 can be independently selected from the group consisting of a substituted or unsubstituted C1 to C20 alkyl group, e.g., methyl or tert-butyl, a substituted or unsubstituted C3 to C20 cycloalkyl, e.g., an adamantanyl unsubstituted or substituted with a phenyl, and a substituted or unsubstituted C6 to C30 aryl group, e.g., phenyl unsubstituted or substituted with C1 to C20 alkyl group. In an aspect of the present disclosure, at least one of d1 to d5 can be a positive integer.

[0173] For example, the third compound **246***a* can be one of the compounds in Formula 6.

##STR00037## ##STR00038## ##STR00039##

[0174] The third compound **246***a* can have a LUMO energy level being in a range of −2.65 eV to −2.55 eV and a HOMO energy level being in a range of −5.55 eV to −5.45 eV.

[0175] A maximum emission wavelength of a mixture of the first and second compounds **242***a* and **244***a* can be shorter than a maximum emission wavelength of the first compound **242***a* and longer than a maximum emission wavelength of the second compound **244***a*.

[0176] In the first compound **242***a* represented by Formula 1, a difference (e.g., a gap) between a singlet energy level and a triplet energy level can be 0.3 eV or less. As a result, the first compound **242***a* can have a delayed fluorescent property so that a non-emissive triplet exciton can be converted into a singlet exciton. Accordingly, the emitting efficiency of the OLED D**1** and the organic light emitting display device **100** including the OLED D**1** can be improved.

[0177] In addition, since the EML **240**A includes the first compound **242***a* as an n-type host and the second compound **244***a* as a p-type host, the exciton generation efficiency in the EML **240**A and an exciton transfer efficiency into the third compound **246***a* can be improved. Accordingly, the emitting efficiency of the OLED D**1** including the first, second and third compounds **242***a*, **244***a* and **246***a* and the organic light emitting display device **100** including the OLED D**1** can be improved.

[0178] Moreover, since the first compound **242***a* represented by formula 1 includes a tetraarylsilane structure represented by Formula 1-1, a boron atom can be protected. Accordingly, the lifespan of the OLED D**1** including the first compound **242***a* and the organic light emitting display device **100** including the OLED D**1** can be improved.

[0179] Further, since a complex formation between the first compound **242***a* represented by formula 1 and the third compound **246***a* represented by formula 5 can be prevented by a tetraarylsilane structure, a concentration of the third compound **246***a* in the EML **240**A can be increased. Accordingly, the emitting efficiency of the OLED D**1** including the first and third compounds **242***a* and **246***a* and the organic light emitting display device **100** including the OLED D**1** can be further improved.

[0180] Referring to FIG. **4**, which is a schematic energy band diagram in an EML of an OLED according to the second embodiment of the present disclosure, a LUMO energy level of the first compound **242***a* is lower than that of the second compound **244***a* and higher than that of the third compound **246***a*. A difference "ΔLUMO_1" between a LUMO energy level of the first compound **242***a* and a LUMO energy level of the third compound **246***a* can be smaller than a difference "ΔLUMO_2" between a LUMO energy level of the first compound **242***a* and a LUMO energy level of the second compound **244***a*.

[0181] For example, difference "ΔLUMO_1" between a LUMO energy level of the first compound **242***a* and a LUMO energy level of the third compound **246***a* can be in a range of 0.03 eV to 0.08 eV, and a difference "ΔLUMO_2" between a LUMO energy level of the first compound **242***a* and a LUMO energy level of the second compound **244***a* can be in a range of 0.10 eV to 0.15 eV.

[0182] A HOMO energy level of the first compound **242***a* is higher than that of the second compound **244***a* and lower than that of the third compound **246***a*. A difference between a HOMO energy level of the first compound **242***a* and a HOMO energy level of the third compound **246***a* can be smaller than a difference between a HOMO energy level of the first compound **242***a* and a HOMO energy level of the second compound **244***a*. For example, a difference between a HOMO energy level of the first compound **242***a* and a HOMO energy level of the third compound **246***a* can be in a range of 0.1 eV to 0.3 eV

[0183] Since the first compound **242***a* represented by Formula 1 has a LUMO energy level higher than that of the third compound **246***a*, an electron trap in the first compound **242***a* can be suppressed. Accordingly, the driving voltage of the OLED D**1** and the organic light emitting display device **100** including the OLED D**1** can be reduced.

[0184] In the EML **240**A, a weight % of each of the first and second compounds **242***a* and **244***a* can be greater than that of the third compound **246***a*. A weight % of the first compound **242***a* and a weight % of the second compound **244***a* can be same or different.

[0185] The first compound **242***a* can have a weight % of 25 to 50, the second compound **244***a* can have a weight % of 25 to 50, and the third compound **246***a* can have a weight % of 5 to 25. In an aspect of the present disclosure, a weight % of the first compound **242***a* and a weight % of the second compound **244***a* can be same, and the third compound **246***a* can have a weight % of 8, 12, 16 or 20.

[0186] In the red pixel region, the OLED includes a red EML. The red EML can include a red host and a red dopant. In the green pixel region, the OLED includes a green EML. The green EML can include a green host and a green dopant. The red dopant can be one of a fluorescent compound, a phosphorescent compound and a delayed fluorescent compound, and the green dopant can be one of a fluorescent compound, a phosphorescent compound and a delayed fluorescent compound.

[0187] As described above, in the blue pixel region, the EML **240**A of the OLED D**1** includes the first compound **242***a* represented by Formula 1, the second compound **244***a* represented by Formula 3 and the third compound **246***a* represented by Formula 5.

[0188] The first compound **242***a* represented by Formula 1 can have a delayed fluorescent property so that the emitting efficiency of the OLED D**1** and the organic light emitting display device **100** including the OLED D**1** can be improved.

[0189] In addition, since the first compound **242***a* represented by formula 1 includes a tetraarylsilane structure represented by Formula 1-1, a boron atom can be protected. Accordingly, the lifespan of the OLED D**1** and the organic light emitting display device **100** including the OLED D**1** can be improved.

[0190] Moreover, since a complex formation between the first compound **242***a* represented by formula 1 and the third compound **246***a* represented by formula 5 can be prevented by a tetraarylsilane structure, a concentration of the third compound **246***a* in the EML **240**A can be increased. Accordingly, the emitting efficiency of the OLED D**1** and the organic light emitting display device **100** including the OLED D**1** can be further improved.

[0191] Furthermore, since the first compound **242***a* represented by Formula 1 has a LUMO energy level higher than that of the third compound **246***a*, an electron trap in the first compound **242***a* can be suppressed. Accordingly, the driving voltage of the OLED D**1** and the organic light emitting display device **100** including the OLED D**1** can be reduced.

[0192] Namely, the OLED D**1** according to the present disclosure and the organic light emitting display device **100** including the OLED D**1** have an advantage in at least one of the driving voltage, the emitting efficiency and the lifespan.

[0193] FIG. **5** is a schematic cross-sectional view of an OLED according to a third embodiment of the present disclosure.

[0194] As shown in FIG. **5**, the OLED D**2** includes first and second electrodes **210** and **230**, which face each other, and an organic light emitting layer **220** therebetween. The organic light emitting layer **220** includes an emitting material layer (EML) **240**B, e.g., a blue EML. The OLED D**2** can further include a capping layer on the second electrode **230** to enhance a light extraction efficiency.

[0195] The organic light emitting display device **100** (of FIG. **2**) can include a red pixel region, a green pixel region and a blue pixel region, and the OLED D**2** can be positioned in the blue pixel region.

[0196] The first electrode **210** can be an anode, and the second electrode **230** can be a cathode. One of the first and second electrodes **210** and **230** can be a reflective electrode, and the other one of the first and second electrodes **210** and **230** can be a transparent (or a semi-transparent) electrode. In the bottom-emission type OLED D**2**, the first electrode **210** can have a single-layered structure of ITO, and the second electrode **230** can be formed of Al. The first electrode **210** can have a thickness of 10 nm to 100 nm, e.g., 40 nm to 60 nm, and the second electrode **230** can have a thickness of 100 nm to 300 nm, e.g., 150 nm to 250 nm.

[0197] The organic light emitting layer **220** can further include at least one of a hole transporting layer (HTL) **260** between the first electrode **210** and the EML **240**B and an electron transporting layer (ETL) **270** between the second electrode **230** and the EML **240**B.

[0198] In addition, the organic light emitting layer **220** can further include at least one of a hole injection layer (HIL) **250** between the first electrode **210** and the HTL **260** and an electron injection layer (EIL) **280** between the second electrode **230** and the ETL **270**.

[0199] Moreover, the organic light emitting layer **220** can further include at least one of an electron blocking layer (EBL) **265** between the HTL **260** and the EML **240**B and a hole blocking layer (HBL) **275** between the EML **240**B and the ETL **270**.

[0200] The HIL **250** can include the above hole injection material and can have a thickness of 1 nm to 20 nm, e.g., 5 nm to 15 nm.

[0201] The HTL **260** can include the above hole transporting material and can have a thickness of 30 nm to 60 nm, e.g., 40 nm to 50 nm.

[0202] The ETL **270** can include the above electron transporting material and can have a thickness of 10 nm to 50 nm, e.g., 20 nm to 40 nm.

[0203] The EIL **280** can include the above electron injection material and can have a thickness of 0.lnm to 10 nm, e.g., lnm to 5 nm.

[0204] The EBL **265** can include the above electron blocking material and can have a thickness of 5 nm to 20 nm, e.g., 5 nm to 15 nm.

[0205] The HBL **275** can include the above hole blocking material and can have a thickness of 5 nm to 20 nm, e.g., 5 nm to 15 nm.

[0206] The EML **240**B can have a thickness of 10 nm to 100 nm, e.g., 10 nm to 50 nm, preferably 20 nm to 40 nm. The EML **240**B can include a first compound **242***b*, a second compound **244***b*, a third compound **246***b* and a fluorescent compound **248***b*. The first compound **242***b* can be an n-type host, the second compound **244***b* can be a p-type host, the third compound **246***b* can be an auxiliary host or an auxiliary dopant, and the fluorescent compound **248***b* can be a dopant (e.g., an emitter).

[0207] The first compound **242***b* is represented by Formula 1 and can be one of the compounds in Formula 2. The second compound **244***b* is represented by Formula 3 and can be one of the

compounds in Formula 4. The third compound **246**_b_ is represented by Formula 5 and can be one of the compounds in Formula 6.
[0208] The fluorescent compound **248**_b_ is represented by Formula 7.
##STR00040##
[0209] In Formula 7, [0210] each of e1 and e6 is independently an integer of 0 to 4, each of e2 to e5 is independently an integer of 0 to 5, [0211] when e1 is 2 or more, two or more R.sub.61 are same or different, when e2 is 2 or more, two or more R.sub.62 are same or different, when e3 is 2 or more, two or more R.sub.63 are same or different, when e4 is 2 or more, two or more R.sub.64 are same or different, when e5 is 2 or more, two or more R.sub.65 are same or different, when e6 is 2 or more, two or more R.sub.66 are same or different, [0212] each of R.sub.61 to R.sub.66 is independently selected from the group consisting of deuterium, a substituted or unsubstituted C1 to C20 alkyl group, a substituted or unsubstituted C1 to C20 alkoxy group, a substituted or unsubstituted C6 to C30 arylsilyl group, a substituted or unsubstituted C6 to C30 arylamino group, a substituted or unsubstituted C6 to C30 aryl group and a substituted or unsubstituted C3 to C30 heteroaryl group, and [0213] each of Ar.sub.4 and Ar.sub.5 is independently selected from the group consisting of a substituted or unsubstituted C6 to C30 arylamino group, a substituted or unsubstituted C6 to C30 aryl group and a substituted or unsubstituted C3 to C30 heteroaryl group.
[0214] In an aspect of the present disclosure, each of Ar.sub.4 and Ar.sub.5 can be independently selected from the group consisting of a substituted or unsubstituted C6 to C30 arylamino group, e.g., diphenylamino, and a substituted or unsubstituted C3 to C30 heteroaryl group, e.g., carbazolyl.
[0215] In an aspect of the present disclosure, each of Ar.sub.4 and Ar.sub.5 can be independently selected from a substituted or unsubstituted C6 to C30 arylamino group, e.g., diphenylamino.
[0216] In an aspect of the present disclosure, each of Ar.sub.4 and Ar.sub.5 can be independently selected from a substituted or unsubstituted C3 to C30 heteroaryl group, e.g., carbazolyl.
[0217] For example, the fluorescent compound **248**_b_ can be one of the compounds in Formula 8.
##STR00041## ##STR00042##
[0218] The fluorescent compound **248**_b_ can have a LUMO energy level being in a range −2.85 eV to −2.75 eV and a HOMO energy level being in a range −5.50 eV to −5.40 eV. In the fluorescent compound **248**_b_, a difference between a singlet energy level and a triplet energy level can be 0.3 eV or less, e.g., 0.01 eV to 0.025 eV. A difference between a singlet energy level and a triplet energy level of the fluorescent compound **248**_b_ can be smaller than a difference between a singlet energy level and a triplet energy level of the first compound **242**_b_.
[0219] A maximum emission wavelength of a mixture of the first and second compounds **242**_b_ and **244**_b_ can be shorter than a maximum emission wavelength of the first compound **242**_b_ and longer than a maximum emission wavelength of the second compound **244**_b_.
[0220] In each of the first compound **242**_b_ represented by Formula 1 and the fluorescent compound **248**_b_ represented by Formula 7, a difference between the singlet energy level and the triplet energy level can be 0.3 eV or less. As a result, each of the first and fluorescent compounds **242**_b_ and **248**_b_ has a delayed fluorescent property so that a non-emissive triplet exciton can be converted into a singlet exciton. Accordingly, the emitting efficiency of the OLED D**2** and the organic light emitting display device **100** including the OLED D**2** can be improved.
[0221] In addition, since the EML **240**B includes the first compound **242**_b_ as an n-type host and the second compound **244**_b_ as a p-type host, the exciton generation efficiency in the EML **240**B and an exciton transfer efficiency into the third compound **246**_b_ and/or the fluorescent compound **248**_b_ can be improved. Accordingly, the emitting efficiency of the OLED D**2** including the first, second, third and fluorescent compounds **242**_b_, **244**_b_, **246**_b_ and **248**_b_ and the organic light emitting display device **100** including the OLED D**2** can be improved.
[0222] Moreover, since the first compound **242**_b_ represented by formula 1 includes a tetraarylsilane structure represented by Formula 1-1, a boron atom can be protected. Accordingly, the lifespan of

the OLED D**2** including the first compound **242***b* and the organic light emitting display device **100** including the OLED D**2** can be improved.
[0223] Further, since a complex formation between the first compound **242***b* represented by formula 1 and the third compound **246***b* represented by formula 5 can be prevented by a tetraarylsilane structure, a concentration of the third compound **246***b* in the EML **240**B can be increased. Accordingly, the emitting efficiency of the OLED D**2** including the first and third compounds **242***b* and **246***b* and the organic light emitting display device **100** including the OLED D**2** can be further improved.
[0224] Referring to FIG. **6**, which is a schematic energy band diagram in an EML of an OLED according to the third embodiment of the present disclosure, a LUMO energy level of the first compound **242***b* is lower than that of the second compound **244***b* and higher than that of the third compound **246***b*. In addition, a LUMO energy level of the third compound **246***b* is higher than that of the fluorescent compound **248***b*.
[0225] A difference "ΔLUMO_1" between a LUMO energy level of the first compound **242***b* and a LUMO energy level of the third compound **246***b* can be smaller than a difference "ΔLUMO_2" between a LUMO energy level of the first compound **242***b* and a LUMO energy level of the second compound **244***b*. In addition, a difference "ΔLUMO_3" between a LUMO energy level of the third compound **246***b* and a LUMO energy level of the fluorescent compound **248***b* can be equal to or greater than a difference "ΔLUMO_2" between a LUMO energy level of the first compound **242***b* and a LUMO energy level of the second compound **244***b*.
[0226] For example, difference "ΔLUMO_1" between a LUMO energy level of the first compound **242***b* and a LUMO energy level of the third compound **246***b* can be in a range of 0.03 eV to 0.08 eV, a difference "ΔLUMO_2" between a LUMO energy level of the first compound **242***b* and a LUMO energy level of the second compound **244***b* can be in a range of 0.10 eV to 0.15 eV, and a difference "ΔLUMO_3" between a LUMO energy level of the third compound **246***b* and a LUMO energy level of the fluorescent compound **248***b* can be in a range of 0.13 eV to 0.25 eV.
[0227] A HOMO energy level of the first compound **242***b* is higher than that of the second compound **244***b* and lower than that of the third compound **246***b*. In addition, a HOMO energy level of the third compound **246***b* can be lower than that of the fluorescent compound **248***b*.
[0228] A difference between a HOMO energy level of the first compound **242***b* and a HOMO energy level of the third compound **246***b* can be smaller than a difference between a HOMO energy level of the first compound **242***b* and a HOMO energy level of the second compound **244***b*. For example, a difference "AHOMO" between a HOMO energy level of the first compound **242***b* and a HOMO energy level of the third compound **246***b* can be in a range of 0.1 eV to 0.3 eV.
[0229] Since the first compound **242***b* represented by Formula 1 has a LUMO energy level higher than that of each of the third and fluorescent compounds **246***b* and **248***b*, an electron trap in the first compound **242***b* can be suppressed. Accordingly, the driving voltage of the OLED D**2** and the organic light emitting display device **100** including the OLED D**2** can be reduced.
[0230] In the EML **240**B, a weight % of each of the first and second compounds **242***b* and **244***b* can be greater than that of each of the third and fluorescent compounds **246***b* and **248***b*. In addition, a weight % of the third compound **246***b* can be greater than that of the fluorescent compound **248***b*. A weight % of the first compound **242***b* and a weight % of the second compound **244***b* can be same or different.
[0231] The first compound **242***b* can have a weight % of 25 to 50, and the second compound **244***b* can have a weight % of 25 to 50. The third compound **246***b* can have a weight % of 5 to 25, and the fluorescent compound **248***b* can have a weight % of 0.1 to 3. In an aspect of the present disclosure, a weight % of the first compound **242***b* and a weight % of the second compound **244***b* can be same, the third compound **246***b* can have a weight % of 8, 12, 16 or 20, and the fluorescent compound **248***b* can have a weight % of 1.
[0232] In the red pixel region, the OLED includes a red EML. The red EML can include a red host

and a red dopant. In the green pixel region, the OLED includes a green EML. The green EML can include a green host and a green dopant. The red dopant can be one of a fluorescent compound, a phosphorescent compound and a delayed fluorescent compound, and the green dopant can be one of a fluorescent compound, a phosphorescent compound and a delayed fluorescent compound.

[0233] As described above, in the blue pixel region, the EML **240**B of the OLED D**2** includes the first compound **242***b* represented by Formula 1, the second compound **244***b* represented by Formula 3, the third compound **246***b* represented by Formula 5, and the fluorescent compound **248***b* represented by Formula 7.

[0234] Each of the first compound **242***b* represented by Formula 1 and the fluorescent compound **248***b* represented by Formula 7 can have a delayed fluorescent property so that the emitting efficiency of the OLED D**2** and the organic light emitting display device **100** including the OLED D**2** can be improved.

[0235] In addition, since the first compound **242***b* represented by formula 1 includes a tetraarylsilane structure represented by Formula 1-1, a boron atom can be protected. Accordingly, the lifespan of the OLED D**2** and the organic light emitting display device **100** including the OLED D**2** can be improved.

[0236] Moreover, since a complex formation between the first compound **242***b* represented by formula 1 and the third compound **246***b* represented by formula 5 can be prevented by a tetraarylsilane structure, a concentration of the third compound **246***b* in the EML **240**B can be increased. Accordingly, the emitting efficiency of the OLED D**2** and the organic light emitting display device **100** including the OLED D**2** can be further improved.

[0237] Furthermore, since the first compound **242***b* represented by Formula 1 has a LUMO energy level higher than that of each of the third and fluorescent compounds **246***b* and **248***b*, an electron trap in the first compound **242***b* can be suppressed. Accordingly, the driving voltage of the OLED D**2** and the organic light emitting display device **100** including the OLED D**2** can be reduced.

[0238] Namely, the OLED D**2** according to the present disclosure and the organic light emitting display device **100** including the OLED D**2** have an advantage in at least one of the driving voltage, the emitting efficiency and the lifespan.

[0239] The EML **240**A in FIG. **3** includes the first, second and third compounds **242***a*, **244***a* and **246***a*, and the EML **240**B in FIG. **5** includes the first, second, third and fluorescent compounds **242***b*, **244***b*, **246***b* and **248***b*. Alternatively, the EML **240**B can include the first, second and fluorescent compounds **242***b*, **244***b* and **248***b* without the third compound **246***b* represented by Formula 5.

[OLED]

[0240] An anode (ITO, 50 nm), an HIL (Formula 9, 7 nm), an HTL (Formula 10, 45 nm), an EBL (Formula 11, 10 nm), an EML (30 nm), an HBL (Formula 12, 10 nm), an ETL (Formula 13, 30 nm), an EIL (LiF, 1 nm) and a cathode (Al, 100 nm) are sequentially deposited to form a blue OLED.

##STR00043## ##STR00044##

1. COMPARATIVE EXAMPLES

(1) Comparative Example 1 (Ref1)

[0241] The compound HH-1 (44 wt %) in Formula 4, the compound EH-B1 (44 wt %) in Formula 14 and the compound PD-1 (12 wt %) in Formula 6 were used to form the EML.

(2) Comparative Example 2 (Ref2)

[0242] The compound HH-1 (42 wt %) in Formula 4, the compound EH-B1 (42 wt %) in Formula 14 and the compound PD-1 (16 wt %) in Formula 6 were used to form the EML.

(3) Comparative Example 3 (Ref3)

[0243] The compound HH-1 (44 wt %) in Formula 4, the compound EH-B2 (44 wt %) in Formula 14 and the compound PD-1 (12 wt %) in Formula 6 were used to form the EML.

(4) Comparative Example 4 (Ref4)

[0244] The compound HH-1 (42 wt %) in Formula 4, the compound EH-B2 (42 wt %) in Formula 14 and the compound PD-1 (16 wt %) in Formula 6 were used to form the EML.

(5) Comparative Example 5 (Ref5)

[0245] The compound HH-1 (44 wt %) in Formula 4, the compound EH-B3 (44 wt %) in Formula 14 and the compound PD-1 (12 wt %) in Formula 6 were used to form the EML.

(6) Comparative Example 6 (Ref6)

[0246] The compound HH-1 (44 wt %) in Formula 4, the compound EH-B4 (44 wt %) in Formula 14 and the compound PD-1 (12 wt %) in Formula 6 were used to form the EML.

##STR00045## ##STR00046##

2. EXAMPLES

(1) Example 1 (Ex1)

[0247] The compound HH-1 (46 wt %) in Formula 4, the compound EH-1 (46 wt %) in Formula 2 and the compound PD-1 (8 wt %) in Formula 6 were used to form the EML.

(2) Example 2 (Ex2)

[0248] The compound HH-1 (44 wt %) in Formula 4, the compound EH-1 (44 wt %) in Formula 2 and the compound PD-1 (12 wt %) in Formula 6 were used to form the EML.

(3) Example 3 (Ex3)

[0249] The compound HH-1 (42 wt %) in Formula 4, the compound EH-1 (42 wt %) in Formula 2 and the compound PD-1 (16 wt %) in Formula 6 were used to form the EML.

(4) Example 4 (Ex4)

[0250] The compound HH-1 (40 wt %) in Formula 4, the compound EH-1 (40 wt %) in Formula 2 and the compound PD-1 (20 wt %) in Formula 6 were used to form the EML.

(5) Example 5 (Ex5)

[0251] The compound HH-1 (46 wt %) in Formula 4, the compound EH-2 (46 wt %) in Formula 2 and the compound PD-1 (8 wt %) in Formula 6 were used to form the EML.

(6) Example 6 (Ex6)

[0252] The compound HH-1 (44 wt %) in Formula 4, the compound EH-2 (44 wt %) in Formula 2 and the compound PD-1 (12 wt %) in Formula 6 were used to form the EML.

(7) Example 7 (Ex7)

[0253] The compound HH-1 (42 wt %) in Formula 4, the compound EH-2 (42 wt %) in Formula 2 and the compound PD-1 (16 wt %) in Formula 6 were used to form the EML.

(8) Example 8 (Ex8)

[0254] The compound HH-1 (40 wt %) in Formula 4, the compound EH-2 (40 wt %) in Formula 2 and the compound PD-1 (20 wt %) in Formula 6 were used to form the EML.

[0255] A LUMO energy level, a HOMO energy level and a difference (ΔE.sub.ST) between a singlet energy level and a triplet energy level of the compounds used in Comparative Examples 1 to 6 and Examples 1 to 8 were measured and listed in Table 1.

[0256] The emitting properties, i.e., a driving voltage (V), an external quantum efficiency (EQE), a color coordinate index (CIEy) and a lifespan (LT(95%)), of the OLED in Comparative Examples 1 to 6 and Examples 1 to 8 were measured at 8.6 mA/cm.sup.2 and listed in Tables 2 and 3. In Tables 2 and 3, the lifespan is a relative value with respect to Comparative Example 1.

TABLE-US-00001 TABLE 1 LUMO (eV) HOMO (eV) ΔE.sub.ST (eV) HH-1 −2.42 −5.82 — EH-1 −2.56 −5.61 0.17 EH-2 −2.54 −5.6 0.19 EH-B1 −3.01 −6.11 0.45 EH-B2 −2.69 −6.05 0.35 EH-B3 −2.62 −5.88 0.2 EH-B4 −2.61 −5.76 0.2 PD-1 −2.6 −5.5 —

TABLE-US-00002 TABLE 2 n-host V EQE (%) CIEy LT (95%) Ref1 EH-B1 4.0 12.5 0.201 100 Ref2 EH-B1 4.1 10.6 0.286 60 Ref3 EH-B2 3.8 14.5 0.188 84 Ref4 EH-B2 4.1 12.2 0.256 40 Ref5 EH-B3 3.9 18.3 0.164 46 Ref6 EH-B4 4 16.2 0.165 55

TABLE-US-00003 TABLE 3 n-host V EQE (%) CIEy LT (95%) Ex1 EH-1 3.9 19.3 0.156 130 Ex2 EH-1 3.8 19.4 0.159 145 Ex3 EH-1 3.7 18.4 0.159 146 Ex4 EH-1 3.8 18.6 0.159 160 Ex5 EH-2 3.7 17.1 0.139 120 Ex6 EH-2 3.6 17.1 0.143 137 Ex7 EH-2 3.5 15.7 0.144 138 Ex8 EH-2 3.5 15.8

0.145 143

[0257] As shown in Tables 2 and 3, in comparison to the OLED of Comparative Examples 1 to 6, the OLED of Examples 1 to 8 has advantages in at least one of the driving voltage, the emitting efficiency, the color purity and the lifespan.

[0258] Referring to Table 1, since each of the compounds EH-B1, EH-B2, EH-B3 and EH-B4 used in the OLED of Comparative Examples 1 to 6 has a LUMO energy level being lower than that of the phosphorescent dopant, i.e., the compound PD-1, the electron is trapped into the n-type host, i.e., the compounds EH-B1, EH-B2, EH-B3 and EH-B4, so that the driving voltage of the OLED of Comparative Examples 1 to 6 is increased. In addition, since a difference (ΔE.sub.ST) between a singlet energy level and a triplet energy level in each of the compounds EH-B1 and EH-B2 is relatively large, each of the compounds EH-B1 and EH-B2 does not provide a delayed fluorescent property so that the emitting efficiency of the OLED of Comparative Examples 1 to 4 is decreased. Moreover, since the n-type host, i.e., the compounds EH-B1, EH-B2, EH-B3 and EH-B4, and the phosphorescent dopant, i.e., the compound PD-1, form a complex, the emitting efficiency of the OLED of Comparative Examples 1 to 6 is decreased.

[0259] On the other hand, since each of the n-type host, i.e., the compound EH-1 or EH-2, used in the OLED of Examples 1 to 8 has a LUMO energy level being higher than that of the phosphorescent dopant, i.e., the compound PD-1, an electron trap in the n-type host can be suppressed or prevented so that the driving voltage of the OLED of Examples 1 to 8 is decreased. In addition, since a difference (ΔE.sub.ST) between a singlet energy level and a triplet energy level in the n-type host, i.e., the compound EH-1 or EH-2, is small, the n-type host, i.e., the compound EH-1 or EH-2, provides a delayed fluorescent property so that the emitting efficiency and the lifespan of the OLED of Examples 1 to 8 are increased. Moreover, a complex generation between the n-type host, i.e., the compound EH-1 or EH-2, and the phosphorescent dopant, i.e., the compound PD-1, can be suppressed or prevented, the emitting efficiency of the OLED of Examples 1 to 8 is further increased. Furthermore, since a boron atom in the n-type host represented by Formula 1, i.e., the compound EH-1 or EH-2, can be protected by a tetraarylsilane moiety, the lifespan of the OLED of Examples 1 to 8 is further increased.

3. COMPARATIVE EXAMPLES

(1) Comparative Example 7 (Ref7)

[0260] The compound HH-1 (43.5 wt %) in Formula 4, the compound EH-B1 (43.5 wt %) in Formula 14, the compound PD-1 (12 wt %) in Formula 6 and the compound FD-1 (1 wt %) in Formula 8 were used to form the EML.

(2) Comparative Example 8 (Ref8)

[0261] The compound HH-1 (41.5 wt %) in Formula 4, the compound EH-B1 (41.5 wt %) in Formula 14, the compound PD-1 (16 wt %) in Formula 6 and the compound FD-1 (1 wt %) in Formula 8 were used to form the EML.

(3) Comparative Example 9 (Ref9)

[0262] The compound HH-1 (43.5 wt %) in Formula 4, the compound EH-B1 (43.5 wt %) in Formula 14, the compound PD-1 (12 wt %) in Formula 6 and the compound FD-2 (1 wt %) in Formula 8 were used to form the EML.

(4) Comparative Example 10 (Ref10)

[0263] The compound HH-1 (43.5 wt %) in Formula 4, the compound EH-B1 (43.5 wt %) in Formula 14, the compound PD-1 (12 wt %) in Formula 6 and the compound FD-3 (1 wt %) in Formula 8 were used to form the EML.

(5) Comparative Example 11 (Ref11)

[0264] The compound HH-1 (43.5 wt %) in Formula 4, the compound EH-B2 (43.5 wt %) in Formula 14, the compound PD-1 (12 wt %) in Formula 6 and the compound FD-1 (1 wt %) in Formula 8 were used to form the EML.

(6) Comparative Example 12 (Ref12)

[0265] The compound HH-1 (41.5 wt %) in Formula 4, the compound EH-B2 (41.5 wt %) in Formula 14, the compound PD-1 (16 wt %) in Formula 6 and the compound FD-1 (1 wt %) in Formula 8 were used to form the EML.

(7) Comparative Example 13 (Ref13)

[0266] The compound HH-1 (43.5 wt %) in Formula 4, the compound EH-B2 (43.5 wt %) in Formula 14, the compound PD-1 (12 wt %) in Formula 6 and the compound FD-2 (1 wt %) in Formula 8 were used to form the EML.

(8) Comparative Example 14 (Ref14)

[0267] The compound HH-1 (43.5 wt %) in Formula 4, the compound EH-B2 (43.5 wt %) in Formula 14, the compound PD-1 (12 wt %) in Formula 6 and the compound FD-3 (1 wt %) in Formula 8 were used to form the EML.

(9) Comparative Example 15 (Ref15)

[0268] The compound HH-1 (41.5 wt %) in Formula 4, the compound EH-B3 (41.5 wt %) in Formula 14, the compound PD-1 (16 wt %) in Formula 6 and the compound FD-1 (1 wt %) in Formula 8 were used to form the EML.

(10) Comparative Example 16 (Ref16)

[0269] The compound HH-1 (43.5 wt %) in Formula 4, the compound EH-B3 (43.5 wt %) in Formula 14, the compound PD-1 (12 wt %) in Formula 6 and the compound FD-2 (1 wt %) in Formula 8 were used to form the EML.

(11) Comparative Example 17 (Ref17)

[0270] The compound HH-1 (43.5 wt %) in Formula 4, the compound EH-B3 (43.5 wt %) in Formula 14, the compound PD-1 (12 wt %) in Formula 6 and the compound FD-3 (1 wt %) in Formula 8 were used to form the EML.

(12) Comparative Example 18 (Ref18)

[0271] The compound HH-1 (41.5 wt %) in Formula 4, the compound EH-B4 (41.5 wt %) in Formula 14, the compound PD-1 (16 wt %) in Formula 6 and the compound FD-1 (1 wt %) in Formula 8 were used to form the EML.

(13) Comparative Example 19 (Ref19)

[0272] The compound HH-1 (43.5 wt %) in Formula 4, the compound EH-B4 (43.5 wt %) in Formula 14, the compound PD-1 (12 wt %) in Formula 6 and the compound FD-2 (1 wt %) in Formula 8 were used to form the EML.

(14) Comparative Example 20 (Ref20)

[0273] The compound HH-1 (43.5 wt %) in Formula 4, the compound EH-B4 (43.5 wt %) in Formula 14, the compound PD-1 (12 wt %) in Formula 6 and the compound FD-3 (1 wt %) in Formula 8 were used to form the EML.

4. EXAMPLES

(1) Example 9 (Ex9)

[0274] The compound HH-1 (45.5 wt %) in Formula 4, the compound EH-1 (45.5 wt %) in Formula 2, the compound PD-1 (8 wt %) in Formula 6, and the compound FD-1 (1 wt %) in Formula 8 were used to form the EML.

(2) Example 10 (Ex10)

[0275] The compound HH-1 (43.5 wt %) in Formula 4, the compound EH-1 (43.5 wt %) in Formula 2, the compound PD-1 (12 wt %) in Formula 6, and the compound FD-1 (1 wt %) in Formula 8 were used to form the EML.

(3) Example 11 (Ex11)

[0276] The compound HH-1 (41.5 wt %) in Formula 4, the compound EH-1 (41.5 wt %) in Formula 2, the compound PD-1 (16 wt %) in Formula 6, and the compound FD-1 (1 wt %) in Formula 8 were used to form the EML.

(4) Example 12 (Ex12)

[0277] The compound HH-1 (39.5 wt %) in Formula 4, the compound EH-1 (39.5 wt %) in

Formula 2, the compound PD-1 (20 wt %) in Formula 6, and the compound FD-1 (1 wt %) in Formula 8 were used to form the EML.

(5) Example 13 (Ex13)

[0278] The compound HH-1 (41.5 wt %) in Formula 4, the compound EH-1 (41.5 wt %) in Formula 2, the compound PD-1 (16 wt %) in Formula 6, and the compound FD-2 (1 wt %) in Formula 8 were used to form the EML.

(6) Example 14 (Ex14)

[0279] The compound HH-1 (39.5 wt %) in Formula 4, the compound EH-1 (39.5 wt %) in Formula 2, the compound PD-1 (20 wt %) in Formula 6, and the compound FD-2 (1 wt %) in Formula 8 were used to form the EML.

(7) Example 15 (Ex15)

[0280] The compound HH-1 (41.5 wt %) in Formula 4, the compound EH-1 (41.5 wt %) in Formula 2, the compound PD-1 (16 wt %) in Formula 6, and the compound FD-3 (1 wt %) in Formula 8 were used to form the EML.

(8) Example 16 (Ex16)

[0281] The compound HH-1 (39.5 wt %) in Formula 4, the compound EH-1 (39.5 wt %) in Formula 2, the compound PD-1 (20 wt %) in Formula 6, and the compound FD-3 (1 wt %) in Formula 8 were used to form the EML.

(9) Example 17 (Ex17)

[0282] The compound HH-1 (45.5 wt %) in Formula 4, the compound EH-2 (45.5 wt %) in Formula 2, the compound PD-1 (8 wt %) in Formula 6, and the compound FD-1 (1 wt %) in Formula 8 were used to form the EML.

(10) Example 18 (Ex18)

[0283] The compound HH-1 (43.5 wt %) in Formula 4, the compound EH-2 (43.5 wt %) in Formula 2, the compound PD-1 (12 wt %) in Formula 6, and the compound FD-1 (1 wt %) in Formula 8 were used to form the EML.

(11) Example 19 (Ex19)

[0284] The compound HH-1 (41.5 wt %) in Formula 4, the compound EH-2 (41.5 wt %) in Formula 2, the compound PD-1 (16 wt %) in Formula 6, and the compound FD-1 (1 wt %) in Formula 8 were used to form the EML.

(12) Example 20 (Ex20)

[0285] The compound HH-1 (39.5 wt %) in Formula 4, the compound EH-2 (39.5 wt %) in Formula 2, the compound PD-1 (20 wt %) in Formula 6, and the compound FD-1 (1 wt %) in Formula 8 were used to form the EML.

(13) Example 21 (Ex21)

[0286] The compound HH-1 (41.5 wt %) in Formula 4, the compound EH-2 (41.5 wt %) in Formula 2, the compound PD-1 (16 wt %) in Formula 6, and the compound FD-2 (1 wt %) in Formula 8 were used to form the EML.

(14) Example 22 (Ex22)

[0287] The compound HH-1 (39.5 wt %) in Formula 4, the compound EH-2 (39.5 wt %) in Formula 2, the compound PD-1 (20 wt %) in Formula 6, and the compound FD-2 (1 wt %) in Formula 8 were used to form the EML.

(14) Example 23 (Ex23)

[0288] The compound HH-1 (41.5 wt %) in Formula 4, the compound EH-2 (41.5 wt %) in Formula 2, the compound PD-1 (16 wt %) in Formula 6, and the compound FD-3 (1 wt %) in Formula 8 were used to form the EML.

(16) Example 24 (Ex24)

[0289] The compound HH-1 (39.5 wt %) in Formula 4, the compound EH-2 (39.5 wt %) in Formula 2, the compound PD-1 (20 wt %) in Formula 6, and the compound FD-3 (1 wt %) in Formula 8 were used to form the EML.

[0290] A LUMO energy level, a HOMO energy level and a difference (ΔE.sub.ST) between a singlet energy level and a triplet energy level of the compounds used in Comparative Examples 7 to 20 and Examples 9 to 24 were measured and listed in Table 4.

[0291] The emitting properties, i.e., a driving voltage (V), an external quantum efficiency (EQE), a color coordinate index (CIEy) and a lifespan (LT(95%)), of the OLED in Comparative Examples 7 to 20 and Examples 9 to 24 were measured at 8.6 mA/cm.sup.2 and listed in Tables 5 to 8. In Tables 5 to 8, the lifespan is a relative value with respect to Comparative Example 1.

TABLE-US-00004 TABLE 4 LUMO (eV) HOMO (eV) ΔE.sub.ST (eV) HH-1 −2.42 −5.82 — EH-1 −2.56 −5.61 0.17 EH-2 −2.54 −5.6 0.19 EH-B1 −3.01 −6.11 0.45 EH-B2 −2.69 −6.05 0.35 EH-B3 −2.62 −5.88 0.2 EH-B4 −2.61 −5.76 0.2 PD-1 −2.6 −5.5 — FD-1 −2.8 −5.4 0.017 FD-2 −2.8 −5.5 0.019 FD-3 −2.8 −5.4 0.018

TABLE-US-00005 TABLE 5 LT n-host FD V EQE (%) CIEy (95%) Ref7 EH-B1 FD-1 4.41 16.2 0.184 40 Ref8 EH-B1 FD-1 4.32 14.1 0.196 41 Ref9 EH-B1 FD-2 4.42 15.2 0.203 20 Ref10 EH-B1 FD-3 4.38 14.3 0.198 32 Ref11 EH-B2 FD-1 4.25 17.3 0.164 80 Ref12 EH-B2 FD-1 4.37 18.1 0.165 39 Ref13 EH-B2 FD-2 4.45 18 0.201 50 Ref14 EH-B2 FD-3 4.42 18.5 0.19 43

TABLE-US-00006 TABLE 6 n-host FD V EQE (%) CIEy LT (95%) Ref15 EH-B3 FD-1 4.2 18.2 0.162 30 Ref16 EH-B3 FD-2 4.32 17.6 0.168 44 Ref17 EH-B3 FD-3 4.11 17.5 0.162 36 Ref18 EH-B4 FD-1 4.2 17.3 0.164 41 Ref19 EH-B4 FD-2 4.3 16.8 0.166 27 Ref20 EH-B4 FD-3 4.13 17.5 0.164 35

TABLE-US-00007 TABLE 7 n-host FD V EQE (%) CIEy LT (95%) Ex9 EH-1 FD-1 3.54 24.7 0.156 142 Ex10 EH-1 FD-1 3.52 24.1 0.157 150 Ex11 EH-1 FD-1 3.47 24 0.156 151 Ex12 EH-1 FD-1 3.41 24.1 0.155 168 Ex13 EH-1 FD-2 3.42 24.3 0.158 165 Ex14 EH-1 FD-2 3.38 25.6 0.159 180 Ex15 EH-1 FD-3 3.41 24.8 0.156 168 Ex16 EH-1 FD-3 3.45 25.1 0.157 178

TABLE-US-00008 TABLE 8 n-host FD V EQE (%) CIEy LT (95%) Ex17 EH-2 FD-1 3.41 23.5 0.155 123 Ex18 EH-2 FD-1 3.45 23.4 0.155 140 Ex19 EH-2 FD-1 3.37 23.1 0.156 146 Ex20 EH-2 FD-1 3.37 23 0.157 153 Ex21 EH-2 FD-2 3.45 24.3 0.158 167 Ex22 EH-2 FD-2 3.44 24.5 0.159 175 Ex23 EH-2 FD-3 3.42 23.8 0.155 170 Ex24 EH-2 FD-3 3.4 24.2 0.156 174

[0292] As shown in Tables 5 and 8, in comparison to the OLED of Comparative Examples 7 to 20, the OLED of Examples 9 to 24 has advantages in the driving voltage, the emitting efficiency, the color purity and the lifespan.

[0293] Referring to Table 4, since each of the compounds EH-B1, EH-B2, EH-B3 and EH-B4 used in the OLED of Comparative Examples 7 to 20 has a LUMO energy level being lower than the phosphorescent dopant, i.e., the compound PD-1, and/or the fluorescent dopant, i.e., the compounds FD-1, FD-2 and FD-3, the electron is trapped into the n-type host, i.e., the compounds EH-B1, EH-B2, EH-B3 and EH-B4, so that the driving voltage of the OLED of Comparative Examples 7 to 20 is increased. In addition, since a difference (ΔE.sub.ST) between a singlet energy level and a triplet energy level in each of the compounds EH-B1 and EH-B2 is relatively large, each of the compounds EH-B1 and EH-B2 does not provide a delayed fluorescent property so that the emitting efficiency of the OLED of Comparative Examples 7 to 14 is decreased. Moreover, since the n-type host, i.e., the compounds EH-B1, EH-B2, EH-B3 and EH-B4, and the phosphorescent dopant, i.e., the compound PD-1, form a complex, the emitting efficiency of the OLED of Comparative Examples 7 to 20 is decreased.

[0294] On the other hand, since each of the n-type host, i.e., the compound EH-1 or EH-2, used in the OLED of Examples 9 to 24 has a LUMO energy level being higher than that of each of the phosphorescent dopant, i.e., the compound PD-1, and the fluorescent dopant, i.e., the compounds FD-1, FD-2 and FD-3, an electron trap in the n-type host can be suppressed or prevented so that the driving voltage of the OLED of Examples 9 to 24 is decreased. In addition, since a difference (ΔE.sub.ST) between a singlet energy level and a triplet energy level in the n-type host, i.e., the compound EH-1 or EH-2, is small, the n-type host, i.e., the compound EH-1 or EH-2, provides a delayed fluorescent property so that the emitting efficiency and the lifespan of the OLED of

Examples 9 to 24 are increased. Moreover, a complex generation between the n-type host, i.e., the compound EH-1 or EH-2, and the phosphorescent dopant, i.e., the compound PD-1, can be suppressed or prevented, the emitting efficiency of the OLED of Examples 9 to 24 is further increased. Furthermore, since a boron atom in the n-type host represented by Formula 1, i.e., the compound EH-1 or EH-2, can be protected by a tetraarylsilane moiety, the lifespan of the OLED of Examples 9 to 24 is further increased.

[0295] FIG. **7** is a schematic cross-sectional view of an OLED according to a fourth embodiment of the present disclosure.

[0296] As illustrated in FIG. **7**, the OLED D**3** includes first and second electrodes **210** and **230** facing each other and an organic light emitting layer **220** therebetween. The organic light emitting layer **220** includes a first emitting part **310** including a first blue EML **320** and a second emitting part **340** including a second blue EML **350**. The organic light emitting layer **220** can include a CGL **370** between the first and second emitting parts **310** and **340**. The OLED D**3** can further include a capping layer on the second electrode **230** to enhance a light extraction efficiency.

[0297] The organic light emitting display device **100** can include a red pixel region, a green pixel region and a blue pixel region, and the OLED D**3** can be positioned in the blue pixel region.

[0298] The first electrode **210** can be an anode, and the second electrode **230** can be a cathode. One of the first and second electrodes **210** and **230** can be a reflective electrode, and the other one of the first and second electrodes **210** and **230** can be a transparent (or a semi-transparent) electrode. For example, the first electrode **210** can have a single-layered structure of ITO, and the second electrode **230** can be formed of Al.

[0299] The first emitting part **310** can further include at least one of a first HTL **334** under the first blue EML **320** and a first ETL **336** over the first blue EML **320**.

[0300] In addition, the first emitting part **310** can further include an HIL **332** between the first electrode **210** and the first HTL **334**.

[0301] Moreover, the first emitting part **310** can further include at least one of a first EBL between the first HTL **334** and the first blue EML **320** and a first HBL between the first blue EML **320** and the first ETL **336**.

[0302] The second emitting part **340** can further include at least one of a second HTL **362** under the second blue EML **350** and a second ETL **364** over the second blue EML **350**.

[0303] In addition, the second emitting part **340** can further include an EIL **366** between the second electrode **230** and the second ETL **364**.

[0304] Moreover, the second emitting part **340** can further include at least one of a second EBL between the second HTL **362** and the second blue EML **350** and a second HBL between the second blue EML **350** and the second ETL **364**.

[0305] For example, the HIL **332** can include the above hole injection material and can have a thickness of 1 nm to 20 nm, e.g., 5 nm to 15 nm.

[0306] Each of the first and second HTLs **334** and **362** can include the above hole transporting material and can have a thickness of 30 nm to 60 nm, e.g., 40 nm to 50 nm.

[0307] Each of the first and second ETLs **336** and **364** can include the above electron transporting material and can have a thickness of 10 nm to 50 nm, e.g., 20 nm to 40 nm.

[0308] The EIL **366** can include the above electron injection material and can have a thickness of 0.1 nm to 10 nm, e.g., 1 nm to 5 nm.

[0309] Each of the first and second EBLs can include the above electron blocking material and can have a thickness of 5 nm to 20 nm, e.g., 5 nm to 15 nm.

[0310] Each of the first and second HBLs can include the above hole blocking material and can have a thickness of 5 nm to 20 nm, e.g., 5 nm to 15 nm.

[0311] The CGL **370** is positioned between the first and second emitting parts **310** and **340**. Namely, the first and second emitting parts **310** and **340** are connected to each other through the CGL **370**. The CGL **370** can be a PN-junction CGL of an N-type CGL **372** and a P-type CGL **374**.

[0312] The N-type CGL **372** is positioned between the first ETL **336** and the second HTL **362**, and the P-type CGL **374** is positioned between the N-type CGL **372** and the second HTL **362**.
[0313] The N-type CGL **372** provides an electron into the first blue EML **320** of the first emitting part **310**, and the P-type CGL **374** provides a hole into the second blue EML **350** of the second emitting part **340**.
[0314] The N-type CGL **372** can be an organic layer doped with an alkali metal, e.g., Li, Na, K and Cs, and/or an alkali earth metal, e.g., Mg, Sr, Ba and Ra. For example, the N-type CGL **372** can be formed of an N-type charge generation material including a host being the organic material, e.g., 4,7-dipheny-1,10-phenanthroline (Bphen) and MTDATA, and a dopant being an alkali metal and/or an alkali earth metal, and the dopant can be doped with a weight % of 0.01 to 30.
[0315] The P-type CGL **374** can be formed of a P-type charge generation material including an inorganic material, e.g., tungsten oxide (WO.sub.x), molybdenum oxide (MoO.sub.x), beryllium oxide (Be.sub.2O.sub.3) or vanadium oxide (V.sub.2O.sub.5), an organic material, e.g., NPD, HAT-CN, F4TCNQ, TPD, TNB, TCTA, N,N′-dioctyl-3,4,9,10-perylenedicarboximide (PTCDI-C8) or their combination.
[0316] The first blue EML **320** can have a thickness in a range of 10 nm to 100 nm, e.g., 10 nm to 50 nm, preferably 20 nm to 40 nm.
[0317] The first blue EML **320** can include a first compound **322**, a second compound **324** and a third compound **326**. The first compound **322** can act as a first host, e.g., an n-type host, the second compound **324** can act as a second host, e.g., a p-type host, and the third compound **326** can act as a dopant.
[0318] The first compound **322** is represented by Formula 1 and can be one of the compounds in Formula 2. The second compound **324** is represented by Formula 3 and can be one of the compounds in Formula 4. The third compound **326** is represented by Formula 5 and can be one of the compounds in Formula 6.
[0319] In the first blue EML **320**, a weight % of each of the first and second compounds **322** and **324** can be greater than that of the third compound **326**. A weight % of the first compound **322** and a weight % of the second compound **324** can be same or different.
[0320] The first compound **322** can have a weight % of 25 to 50, the second compound **324** can have a weight % of 25 to 50, and the third compound **326** can have a weight % of 5 to 25. In an aspect of the present disclosure, a weight % of the first compound **322** and a weight % of the second compound **324** can be same, and the third compound **326** can have a weight % of 8, 12, 16 or 20.
[0321] In an aspect of the present disclosure, the first blue EML **320** can include a first compound **322**, a second compound **324**, a third compound **326** and a fluorescent compound. The first compound **322** can act as a first host, e.g., an n-type host, and the second compound **324** can act as a second host, e.g., a p-type host. The third compound **326** can act as an auxiliary host or an auxiliary dopant, and the fluorescent compound can act as a dopant.
[0322] The first compound **322** is represented by Formula 1 and can be one of the compounds in Formula 2. The second compound **324** is represented by Formula 3 and can be one of the compounds in Formula 4. The third compound **326** is represented by Formula 5 and can be one of the compounds in Formula 6. The fluorescent compound is represented by Formula 7 and can be one of the compounds in Formula 8.
[0323] In the first blue EML **320**, a weight % of each of the first and second compounds **322** and **324** can be greater than that of each of the third compound **326** and the fluorescent compound. In addition, a weight % of the third compound **326** can be greater than that of the fluorescent compound. A weight % of the first compound **322** and a weight % of the second compound **324** can be same or different.
[0324] The first compound **322** can have a weight % of 25 to 50, and the second compound **324** can have a weight % of 25 to 50. The third compound **326** can have a weight % of 5 to 25, and the

fluorescent compound can have a weight % of 0.1 to 3. In an aspect of the present disclosure, a weight % of the first compound **322** and a weight % of the second compound **324** can be same, the third compound **326** can have a weight % of 8, 12, 16 or 20, and the fluorescent compound can have a weight % of 1.

[0325] The second blue EML **350** includes a compound **352** as a blue host and a compound **354** as a blue dopant. For example, the compound **354** can be one of a blue fluorescent dopant, a blue phosphorescent dopant and a blue delayed fluorescent dopant. The second blue EML **350** can have a thickness in a range of 10 nm to 100 nm, e.g., 20 nm to 40 nm.

[0326] For example, the compound **352** can be selected from the group consisting of mCP, 9-(3-(9H-carbazol-9-yl)phenyl)-9H-carbazole-3-carbonitrile (mCP-CN), mCBP, CBP-CN, 9-(3-(9H-Carbazol-9-yl)phenyl)-3-(diphenylphosphoryl)-9H-carbazole (mCPPO1), 3,5-Di(9H-carbazol-9-yl)biphenyl (Ph-mCP), TSPO1, 9-(3′-(9H-carbazol-9-yl)-[1,1′-biphenyl]-3-yl)-9H-pyrido[2,3-b]indole (CzBPCb), bis(2-methylphenyl)diphenylsilane (UGH-1), 1,4-bis(triphenylsilyl)benzene (UGH-2), 1,3-bis(triphenylsilyl)benzene (UGH-3), 9,9-spirobifluoren-2-yl-diphenyl-phosphine oxide (SPPO1), and 9,9′-(5-(triphenylsilyl)-1,3-phenylene)bis(9H-carbazole) (SimCP).

[0327] For example, the compound **354** can be selected from the group consisting of perylene, 4,4′-bis[4-(di-p-tolylamino)styryl]biphenyl (DPAVBi), 4-(di-p-tolylamino)-4-4′-[(di-p-tolylamino)styryl]stilbene (DPAVB), 4,4′-bis[4-(diphenylamino)styryl]biphenyl (BDAVBi), 2,7-bis(4-diphenylamino)styryl-9,9-spirofluorene (spiro-DPVBi), 1,4-bis[2-[4-[N,N-di(p-tolyl)amino]phenyl]vinyl]benzene (DSB), 1-4-di-[4-(N,N-diphenyl)amino]styryl-benzene (DSA), 2,5,8,11-tetra-tert-butylperylene (TBPe), (bis(2-hydroxylphenyl)-pyridine)beryllium (Bepp2) and 9-(9-Phenylcarbazole-3-yl)-10-(naphthalene-1-yl)anthracene (PCAN).

[0328] In the blue pixel region, the organic light emitting layer **220** of the OLED D**3** includes the first blue EML **320** and the second blue EML **350** to have a tandem structure.

[0329] In this case, the first blue EML **320** can include the first compound **322** represented by Formula 1, the second compound **324** represented by Formula 3 and the third compound **326** represented by Formula 5.

[0330] In the first compound **322** represented by Formula 1, a difference between a singlet energy level and a triplet energy level can be 0.3 eV or less. As a result, the first compound **322** can have a delayed fluorescent property so that a non-emissive triplet exciton can be converted into a singlet exciton. Accordingly, the emitting efficiency of the OLED D**3** and the organic light emitting display device **100** including the OLED D**3** can be improved.

[0331] In addition, since the first blue EML **320** includes the first compound **322** as an n-type host and the second compound **324** as a p-type host, the exciton generation efficiency in the first blue EML **320** and an exciton transfer efficiency into the third compound **326** can be improved. Accordingly, the emitting efficiency of the OLED D**3** including the first, second and third compounds **322**, **324** and **326** and the organic light emitting display device **100** including the OLED D**3** can be improved.

[0332] Moreover, since the first compound **322** represented by formula 1 includes a tetraarylsilane structure represented by Formula 1-1, a boron atom can be protected. Accordingly, the lifespan of the OLED D**3** including the first compound **322** and the organic light emitting display device **100** including the OLED D**3** can be improved.

[0333] Further, since a complex formation between the first compound **322** represented by formula 1 and the third compound **326** represented by formula 5 can be prevented by a tetraarylsilane structure, a concentration of the third compound **326** in the first blue EML **320** can be increased. Accordingly, the emitting efficiency of the OLED D**3** including the first and third compounds **322** and **326** and the organic light emitting display device **100** including the OLED D**3** can be further improved.

[0334] Furthermore, since the first compound **322** represented by Formula 1 has a LUMO energy level higher than that of the third compound **326**, an electron trap in the first compound **322** can be suppressed. Accordingly, the driving voltage of the OLED D**3** and the organic light emitting display device **100** including the OLED D**3** can be reduced.

[0335] In an aspect of the present disclosure, the first blue EML **320** can include the first compound **322** represented by Formula 1, the second compound represented by Formula 3, the third compound represented by Formula 5 and the fluorescent compound represented by Formula 7.

[0336] In each of the first compound **322** represented by Formula 1 and the fluorescent compound represented by Formula 7, a difference between a singlet energy level and a triplet energy level can be 0.3 eV or less. As a result, each of the first compound **322** and the fluorescent compound can have a delayed fluorescent property so that a non-emissive triplet exciton can be converted into a singlet exciton. Accordingly, the emitting efficiency of the OLED D**3** and the organic light emitting display device **100** including the OLED D**3** can be improved.

[0337] In addition, since the first blue EML **320** includes the first compound **322** as an n-type host and the second compound **324** as a p-type host, the exciton generation efficiency in the first blue EML **320** and an exciton transfer efficiency into the third compound **326** and/or the fluorescent compound can be improved. Accordingly, the emitting efficiency of the OLED D**3** including the first, second and third compounds **322**, **324** and **326** and the fluorescent compound and the organic light emitting display device **100** including the OLED D**3** can be improved.

[0338] Moreover, since the first compound **322** represented by formula 1 includes a tetraarylsilane structure represented by Formula 1-1, a boron atom can be protected. Accordingly, the lifespan of the OLED D**3** including the first compound **322** and the organic light emitting display device **100** including the OLED D**3** can be improved.

[0339] Further, since a complex formation between the first compound **322** represented by formula 1 and the third compound **326** represented by formula 5 can be prevented by a tetraarylsilane structure, a concentration of the third compound **326** in the first blue EML **320** can be increased. Accordingly, the emitting efficiency of the OLED D**3** including the first and third compounds **322** and **326** and the organic light emitting display device **100** including the OLED D**3** can be further improved.

[0340] Furthermore, since the first compound **322** represented by Formula 1 has a LUMO energy level higher than that of each of the third compound **326** and the fluorescent compound, an electron trap in the first compound **322** can be suppressed. Accordingly, the driving voltage of the OLED D**3** and the organic light emitting display device **100** including the OLED D**3** can be reduced.

[0341] FIG. **8** is a schematic cross-sectional view of an OLED according to a fifth embodiment of the present disclosure.

[0342] As illustrated in FIG. **8**, the OLED D**4** includes first and second electrodes **210** and **230** facing each other and an organic light emitting layer **220** therebetween. The organic light emitting layer **220** includes a first emitting part **410** including a first blue EML **420** and a second emitting part **440** including a second blue EML **450**. The organic light emitting layer **220** can include a CGL **470** between the first and second emitting parts **410** and **440**. The OLED D**4** can further include a capping layer on the second electrode **230** to enhance a light extraction efficiency.

[0343] The organic light emitting display device **100** can include a red pixel region, a green pixel region and a blue pixel region, and the OLED D**4** can be positioned in the blue pixel region.

[0344] The first electrode **210** can be an anode, and the second electrode **230** can be a cathode. One of the first and second electrodes **210** and **230** can be a reflective electrode, and the other one of the first and second electrodes **210** and **230** can be a transparent (or a semi-transparent) electrode. For example, the first electrode **210** can have a single-layered structure of ITO, and the second electrode **230** can be formed of Al.

[0345] The first emitting part **410** can further include at least one of a first HTL **434** under the first blue EML **420** and a first ETL **436** over the first blue EML **420**.

[0346] In addition, the first emitting part **410** can further include an HIL **432** between the first electrode **210** and the first HTL **434**.

[0347] Moreover, the first emitting part **410** can further include at least one of a first EBL between the first HTL **434** and the first blue EML **420** and a first HBL between the first blue EML **420** and the first ETL **436**.
[0348] The second emitting part **440** can further include at least one of a second HTL **462** under the second blue EML **450** and a second ETL **464** over the second blue EML **450**.
[0349] In addition, the second emitting part **440** can further include an EIL **466** between the second electrode **230** and the second ETL **464**.
[0350] Moreover, the second emitting part **440** can further include at least one of a second EBL between the second HTL **462** and the second blue EML **450** and a second HBL between the second blue EML **450** and the second ETL **464**.
[0351] For example, the HIL **432** can include the above hole injection material and can have a thickness of 1 nm to 20 nm, e.g., 5 nm to 15 nm.
[0352] Each of the first and second HTLs **434** and **462** can include the above hole transporting material and can have a thickness of 30 nm to 60 nm, e.g., 40 nm to 50 nm.
[0353] Each of the first and second ETLs **436** and **464** can include the above electron transporting material and can have a thickness of 10 nm to 50 nm, e.g., 20 nm to 40 nm.
[0354] The EIL **466** can include the above electron injection material and can have a thickness of 0.1 nm to 10 nm, e.g., 1 nm to 5 nm.
[0355] Each of the first and second EBLs can include the above electron blocking material and can have a thickness of 5 nm to 20 nm, e.g., 5 nm to 15 nm.
[0356] Each of the first and second HBLs can include the above hole blocking material and can have a thickness of 5 nm to 20 nm, e.g., 5 nm to 15 nm.
[0357] The CGL **470** is positioned between the first and second emitting parts **410** and **440**. Namely, the first and second emitting parts **410** and **440** are connected to each other through the CGL **470**. The CGL **470** can be a PN-junction CGL of an N-type CGL **472** and a P-type CGL **474**.
[0358] The N-type CGL **472** is positioned between the first ETL **436** and the second HTL **462**, and the P-type CGL **474** is positioned between the N-type CGL **472** and the second HTL **462**.
[0359] The N-type CGL **472** provides an electron into the first blue EML **420** of the first emitting part **410**, and the P-type CGL **474** provides a hole into the second blue EML **450** of the second emitting part **440**.
[0360] The N-type CGL **472** can be an organic layer doped with an alkali metal, e.g., Li, Na, K and Cs, and/or an alkali earth metal, e.g., Mg, Sr, Ba and Ra. For example, the N-type CGL **472** can be formed of an N-type charge generation material including a host being the organic material, e.g., 4,7-dipheny-1,10-phenanthroline (Bphen) and MTDATA, and a dopant being an alkali metal and/or an alkali earth metal, and the dopant can be doped with a weight % of 0.01 to 30.
[0361] The P-type CGL **474** can be formed of a P-type charge generation material including an inorganic material, e.g., tungsten oxide (WO.sub.x), molybdenum oxide (MoO.sub.x), beryllium oxide (Be.sub.2O.sub.3) or vanadium oxide (V.sub.2O.sub.5), an organic material, e.g., NPD, HAT-CN, F4TCNQ, TPD, TNB, TCTA, N,N′-dioctyl-3,4,9,10-perylenedicarboximide (PTCDI-C8) or their combination.
[0362] The first blue EML **420** includes a compound **422** as a blue host and a compound **424** as a blue dopant. For example, the compound **424** can be one of a blue fluorescent dopant, a blue phosphorescent dopant and a blue delayed fluorescent dopant. The first blue EML **420** can have a thickness in a range of 10 nm to 100 nm, e.g., 20 nm to 40 nm.
[0363] For example, the blue host can be one of the above blue host materials, and the blue dopant can be one of the above blue dopant materials.
[0364] The second blue EML **450** can have a thickness in a range of 10 nm to 100 nm, e.g., 10 nm to 50 nm, preferably 20 nm to 40 nm.
[0365] The second blue EML **450** can include a first compound **452**, a second compound **454** and a third compound **456**. The first compound **452** can act as a first host, e.g., an n-type host, the second

compound **454** can act as a second host, e.g., a p-type host, and the third compound **456** can act as a dopant.
[0366] The first compound **452** is represented by Formula 1 and can be one of the compounds in Formula 2. The second compound **454** is represented by Formula 3 and can be one of the compounds in Formula 4. The third compound **456** is represented by Formula 5 and can be one of the compounds in Formula 6.
[0367] In the second blue EML **450**, a weight % of each of the first and second compounds **452** and **454** can be greater than that of the third compound **456**. A weight % of the first compound **452** and a weight % of the second compound **454** can be same or different.
[0368] The first compound **452** can have a weight % of 25 to 50, the second compound **454** can have a weight % of 25 to 50, and the third compound **456** can have a weight % of 5 to 25. In an aspect of the present disclosure, a weight % of the first compound **452** and a weight % of the second compound **454** can be same, and the third compound **456** can have a weight % of 8, 12, 16 or 20.
[0369] In an aspect of the present disclosure, the second blue EML **450** can include a first compound **452**, a second compound **454**, a third compound **456** and a fluorescent compound. The first compound **452** can act as a first host, e.g., an n-type host, and the second compound **454** can act as a second host, e.g., a p-type host. The third compound **456** can act as an auxiliary host or an auxiliary dopant, and the fluorescent compound can act as a dopant.
[0370] The first compound **452** is represented by Formula 1 and can be one of the compounds in Formula 2. The second compound **454** is represented by Formula 3 and can be one of the compounds in Formula 4. The third compound **456** is represented by Formula 5 and can be one of the compounds in Formula 6. The fluorescent compound is represented by Formula 7 and can be one of the compounds in Formula 8.
[0371] In the second blue EML **450**, a weight % of each of the first and second compounds **452** and **454** can be greater than that of each of the third compound **456** and the fluorescent compound. In addition, a weight % of the third compound **456** can be greater than that of the fluorescent compound. A weight % of the first compound **452** and a weight % of the second compound **454** can be same or different.
[0372] The first compound **452** can have a weight % of 25 to 50, and the second compound **454** can have a weight % of 25 to 50. The third compound **456** can have a weight % of 5 to 25, and the fluorescent compound can have a weight % of 0.1 to 3. In an aspect of the present disclosure, a weight % of the first compound **452** and a weight % of the second compound **454** can be same, the third compound **456** can have a weight % of 8, 12, 16 or 20, and the fluorescent compound can have a weight % of 1.
[0373] In the blue pixel region, the organic light emitting layer **220** of the OLED D**4** includes the first blue EML **420** and the second blue EML **450** to have a tandem structure.
[0374] In this case, the second blue EML **450** can include the first compound **452** represented by Formula 1, the second compound **454** represented by Formula 3 and the third compound **456** represented by Formula 5.
[0375] In the first compound **452** represented by Formula 1, a difference between a singlet energy level and a triplet energy level can be 0.3 eV or less. As a result, the first compound **452** can have a delayed fluorescent property so that a non-emissive triplet exciton can be converted into a singlet exciton. Accordingly, the emitting efficiency of the OLED D**4** and the organic light emitting display device **100** including the OLED D**4** can be improved.
[0376] In addition, since the second blue EML **450** includes the first compound **452** as an n-type host and the second compound **454** as a p-type host, the exciton generation efficiency in the second blue EML **450** and an exciton transfer efficiency into the third compound **456** can be improved. Accordingly, the emitting efficiency of the OLED D**4** including the first, second and third compounds **452**, **454** and **456** and the organic light emitting display device **100** including the

OLED D**4** can be improved.

[0377] Moreover, since the first compound **452** represented by formula 1 includes a tetraarylsilane structure represented by Formula 1-1, a boron atom can be protected. Accordingly, the lifespan of the OLED D**4** including the first compound **452** and the organic light emitting display device **100** including the OLED D**4** can be improved.

[0378] Further, since a complex formation between the first compound **452** represented by formula 1 and the third compound **456** represented by formula 5 can be prevented by a tetraarylsilane structure, a concentration of the third compound **456** in the second blue EML **450** can be increased. Accordingly, the emitting efficiency of the OLED D**4** including the first and third compounds **452** and **456** and the organic light emitting display device **100** including the OLED D**4** can be further improved.

[0379] Furthermore, since the first compound **452** represented by Formula 1 has a LUMO energy level higher than the third compound **456**, an electron trap in the first compound **452** can be suppressed. Accordingly, the driving voltage of the OLED D**4** and the organic light emitting display device **100** including the OLED D**4** can be reduced.

[0380] In an aspect of the present disclosure, the second blue EML **450** can include the first compound **452** represented by Formula 1, the second compound represented by Formula 3, the third compound represented by Formula 5 and the fluorescent compound represented by Formula 7.

[0381] In each of the first compound **452** represented by Formula 1 and the fluorescent compound represented by Formula 7, a difference between a singlet energy level and a triplet energy level can be 0.3 eV or less. As a result, each of the first compound **452** and the fluorescent compound can have a delayed fluorescent property so that a non-emissive triplet exciton can be converted into a singlet exciton. Accordingly, the emitting efficiency of the OLED D**4** and the organic light emitting display device **100** including the OLED D**4** can be improved.

[0382] In addition, since the second blue EML **450** includes the first compound **452** as an n-type host and the second compound **454** as a p-type host, the exciton generation efficiency in the second blue EML **450** and an exciton transfer efficiency into the third compound **456** and/or the fluorescent compound can be improved. Accordingly, the emitting efficiency of the OLED D**4** including the first, second and third compounds **452**, **454** and **456** and the fluorescent compound and the organic light emitting display device **100** including the OLED D**4** can be improved.

[0383] Moreover, since the first compound **452** represented by formula 1 includes a tetraarylsilane structure represented by Formula 1-1, a boron atom can be protected. Accordingly, the lifespan of the OLED D**4** including the first compound **452** and the organic light emitting display device **100** including the OLED D**4** can be improved.

[0384] Further, since a complex formation between the first compound **452** represented by formula 1 and the third compound **456** represented by formula 5 can be prevented by a tetraarylsilane structure, a concentration of the third compound **456** in the second blue EML **450** can be increased. Accordingly, the emitting efficiency of the OLED D**4** including the first and third compounds **452** and **456** and the organic light emitting display device **100** including the OLED D**4** can be further improved.

[0385] Furthermore, since the first compound **452** represented by Formula 1 has a LUMO energy level higher than that of each of the third compound **456** and the fluorescent compound, an electron trap in the first compound **452** can be suppressed. Accordingly, the driving voltage of the OLED D**4** and the organic light emitting display device **100** including the OLED D**4** can be reduced.

[0386] FIG. **9** is a schematic cross-sectional view of an OLED according to a sixth embodiment of the present disclosure.

[0387] As illustrated in FIG. **9**, the OLED D**5** includes first and second electrodes **210** and **230** facing each other and an organic light emitting layer **220** therebetween. The organic light emitting layer **220** includes a first emitting part **510** including a first blue EML **520** and a second emitting part **540** including a second blue EML **550**. The organic light emitting layer **220** can include a CGL

570 between the first and second emitting parts **510** and **540**. The OLED D**5** can further include a capping layer on the second electrode **230** to enhance a light extraction efficiency.
[0388] The organic light emitting display device **100** can include a red pixel region, a green pixel region and a blue pixel region, and the OLED D**5** can be positioned in the blue pixel region.
[0389] The first electrode **210** can be an anode, and the second electrode **230** can be a cathode. One of the first and second electrodes **210** and **230** can be a reflective electrode, and the other one of the first and second electrodes **210** and **230** can be a transparent (or a semi-transparent) electrode. For example, the first electrode **210** can have a single-layered structure of ITO, and the second electrode **230** can be formed of Al.
[0390] The first emitting part **510** can further include at least one of a first HTL **534** under the first blue EML **520** and a first ETL **536** over the first blue EML **520**.
[0391] In addition, the first emitting part **510** can further include an HIL **532** between the first electrode **210** and the first HTL **534**.
[0392] Moreover, the first emitting part **510** can further include at least one of a first EBL between the first HTL **534** and the first blue EML **520** and a first HBL between the first blue EML **520** and the first ETL **536**.
[0393] The second emitting part **540** can further include at least one of a second HTL **562** under the second blue EML **550** and a second ETL **564** over the second blue EML **550**.
[0394] In addition, the second emitting part **540** can further include an EIL **566** between the second electrode **230** and the second ETL **564**.
[0395] Moreover, the second emitting part **540** can further include at least one of a second EBL between the second HTL **562** and the second blue EML **550** and a second HBL between the second blue EML **550** and the second ETL **564**.
[0396] For example, the HIL **532** can include the above hole injection material and can have a thickness of 1 nm to 20 nm, e.g., 5 nm to 15 nm.
[0397] Each of the first and second HTLs **534** and **562** can include the above hole transporting material and can have a thickness of 30 nm to 60 nm, e.g., 40 nm to 50 nm.
[0398] Each of the first and second ETLs **536** and **564** can include the above electron transporting material and can have a thickness of 10 nm to 50 nm, e.g., 20 nm to 40 nm.
[0399] The EIL **566** can include the above electron injection material and can have a thickness of 0.1 nm to 10 nm, e.g., 1 nm to 5 nm.
[0400] Each of the first and second EBLs can include the above electron blocking material and can have a thickness of 5 nm to 20 nm, e.g., 5 nm to 15 nm.
[0401] Each of the first and second HBLs can include the above hole blocking material and can have a thickness of 5 nm to 20 nm, e.g., 5 nm to 15 nm.
[0402] The CGL **570** is positioned between the first and second emitting parts **510** and **540**. Namely, the first and second emitting parts **510** and **540** are connected to each other through the CGL **570**. The CGL **570** can be a PN-junction CGL of an N-type CGL **572** and a P-type CGL **574**.
[0403] The N-type CGL **572** is positioned between the first ETL **536** and the second HTL **562**, and the P-type CGL **574** is positioned between the N-type CGL **572** and the second HTL **562**.
[0404] The N-type CGL **572** provides an electron into the first blue EML **520** of the first emitting part **510**, and the P-type CGL **574** provides a hole into the second blue EML **550** of the second emitting part **540**.
[0405] The N-type CGL **572** can be an organic layer doped with an alkali metal, e.g., Li, Na, K and Cs, and/or an alkali earth metal, e.g., Mg, Sr, Ba and Ra. For example, the N-type CGL **572** can be formed of an N-type charge generation material including a host being the organic material, e.g., 4,7-dipheny-1,10-phenanthroline (Bphen) and MTDATA, and a dopant being an alkali metal and/or an alkali earth metal, and the dopant can be doped with a weight % of 0.01 to 30.
[0406] The P-type CGL **574** can be formed of a P-type charge generation material including an inorganic material, e.g., tungsten oxide (WO.sub.x), molybdenum oxide (MoO.sub.x), beryllium

oxide (Be.sub.2O.sub.3) or vanadium oxide (V.sub.2O.sub.5), an organic material, e.g., NPD, HAT-CN, F4TCNQ, TPD, TNB, TCTA, N,N′-dioctyl-3,4,9,10-perylenedicarboximide (PTCDI-C8) or their combination.

[0407] The first blue EML **520** can have a thickness in a range of 10 nm to 100 nm, e.g., 10 nm to 50 nm, preferably 20 nm to 40 nm.

[0408] The first blue EML **520** can include a first compound **522**, a second compound **524** and a third compound **526**. The first compound **522** can act as a first host, e.g., an n-type host, the second compound **524** can act as a second host, e.g., a p-type host, and the third compound **526** can act as a dopant.

[0409] The first compound **522** is represented by Formula 1 and can be one of the compounds in Formula 2. The second compound **524** is represented by Formula 3 and can be one of the compounds in Formula 4. The third compound **526** is represented by Formula 5 and can be one of the compounds in Formula 6.

[0410] In the first blue EML **520**, a weight % of each of the first and second compounds **522** and **524** can be greater than that of the third compound **526**. A weight % of the first compound **522** and a weight % of the second compound **524** can be same or different.

[0411] The first compound **522** can have a weight % of 25 to 50, the second compound **524** can have a weight % of 25 to 50, and the third compound **526** can have a weight % of 5 to 25. In an aspect of the present disclosure, a weight % of the first compound **522** and a weight % of the second compound **524** can be same, and the third compound **526** can have a weight % of 8, 12, 16 or 20.

[0412] In an aspect of the present disclosure, the first blue EML **520** can include a first compound **522**, a second compound **524**, a third compound **526** and a first fluorescent compound. The first compound **522** can act as a first host, e.g., an n-type host, and the second compound **524** can act as a second host, e.g., a p-type host. The third compound **526** can act as an auxiliary host or an auxiliary dopant, and the first fluorescent compound can act as a dopant.

[0413] The first compound **522** is represented by Formula 1 and can be one of the compounds in Formula 2. The second compound **524** is represented by Formula 3 and can be one of the compounds in Formula 4. The third compound **526** is represented by Formula 5 and can be one of the compounds in Formula 6. The first fluorescent compound is represented by Formula 7 and can be one of the compounds in Formula 8.

[0414] In the first blue EML **520**, a weight % of each of the first and second compounds **522** and **524** can be greater than that of each of the third compound **526** and the first fluorescent compound. In addition, a weight % of the third compound **526** can be greater than that of the first fluorescent compound. A weight % of the first compound **522** and a weight % of the second compound **524** can be same or different.

[0415] The first compound **522** can have a weight % of 25 to 50, and the second compound **524** can have a weight % of 25 to 50. The third compound **526** can have a weight % of 5 to 25, and the first fluorescent compound can have a weight % of 0.1 to 3. In an aspect of the present disclosure, a weight % of the first compound **522** and a weight % of the second compound **524** can be same, the third compound **526** can have a weight % of 8, 12, 16 or 20, and the first fluorescent compound can have a weight % of 1.

[0416] The second blue EML **550** can have a thickness in a range of 10 nm to 100 nm, e.g., 10 nm to 50 nm, preferably 20 nm to 40 nm.

[0417] The second blue EML **550** can include a fourth compound **552**, a fifth compound **554** and a sixth compound **556**. The fourth compound **552** can act as a first host, e.g., an n-type host, the fifth compound **554** can act as a second host, e.g., a p-type host, and the sixth compound **556** can act as a dopant.

[0418] The fourth compound **552** is represented by Formula 1 and can be one of the compounds in Formula 2. The fifth compound **554** is represented by Formula 3 and can be one of the compounds in Formula 4. The sixth compound **556** is represented by Formula 5 and can be one of the compounds in Formula 6.

[0419] In the second blue EML **550**, a weight % of each of the fourth and fifth compounds **552** and **554** can be greater than that of the sixth compound **556**. A weight % of the fourth compound **552** and a weight % of the fifth compound **554** can be same or different.

[0420] The fourth compound **552** can have a weight % of 25 to 50, the fifth compound **554** can have a weight % of 25 to 50, and the sixth compound **556** can have a weight % of 5 to 25. In an aspect of the present disclosure, a weight % of the fourth compound **552** and a weight % of the fifth compound **554** can be same, and the sixth compound **556** can have a weight % of 8, 12, 16 or 20.

[0421] In an aspect of the present disclosure, the second blue EML **550** can include a fourth compound **552**, a fifth compound **554**, a sixth compound **556** and a second fluorescent compound. The fourth compound **552** can act as a first host, e.g., an n-type host, and the fifth compound **554** can act as a second host, e.g., a p-type host. The sixth compound **556** can act as an auxiliary host or an auxiliary dopant, and the second fluorescent compound can act as a dopant.

[0422] The fourth compound **552** is represented by Formula 1 and can be one of the compounds in Formula 2. The fifth compound **554** is represented by Formula 3 and can be one of the compounds in Formula 4. The sixth compound **556** is represented by Formula 5 and can be one of the compounds in Formula 6. The second fluorescent compound is represented by Formula 7 and can be one of the compounds in Formula 8.

[0423] In the second blue EML **550**, a weight % of each of the fourth and fifth compounds **552** and **554** can be greater than that of each of the sixth compound **556** and the second fluorescent compound. In addition, a weight % of the sixth compound **556** can be greater than that of the second fluorescent compound. A weight % of the fourth compound **552** and a weight % of the fifth compound **554** can be same or different.

[0424] The fourth compound **552** can have a weight % of 25 to 50, and the fifth compound **554** can have a weight % of 25 to 50. The sixth compound **556** can have a weight % of 5 to 25, and the second fluorescent compound can have a weight % of 0.1 to 3. In an aspect of the present disclosure, a weight % of the fourth compound **552** and a weight % of the fifth compound **554** can be same, the sixth compound **556** can have a weight % of 8, 12, 16 or 20, and the second fluorescent compound can have a weight % of 1.

[0425] The first compound **522** in the first blue EML **520** and the fourth compound **552** in the second blue EML **550** can be same or different. The second compound **524** in the first blue EML **520** and the fifth compound **554** in the second blue EML **550** can be same or different. The third compound **526** in the first blue EML **520** and the sixth compound **556** in the second blue EML **550** can be same or different. The first fluorescent compound in the first blue EML **520** and the second fluorescent compound in the second blue EML **550** can be same or different.

[0426] A weight % of the first compound **522** in the first blue EML **520** and a weight % of the fourth compound **552** in the second blue EML **550** can be same or different. A weight % of the second compound **524** in the first blue EML **520** and a weight % of the fifth compound **554** in the second blue EML **550** can be same or different. A weight % of the third compound **526** in the first blue EML **520** and a weight % of the sixth compound **556** in the second blue EML **550** can be same or different. A weight % of the first fluorescent compound in the first blue EML **520** and a weight % of the second fluorescent compound in the second blue EML **550** can be same or different.

[0427] In the blue pixel region, the organic light emitting layer **220** of the OLED D**5** includes the first blue EML **520** and the second blue EML **550** to have a tandem structure.

[0428] In this case, the first blue EML **520** can include the first compound **522** represented by Formula 1, the second compound **524** represented by Formula 3 and the third compound **526** represented by Formula 5, and the second blue EML **550** can include the fourth compound **552** represented by Formula 1, the fifth compound **554** represented by Formula 3 and the sixth compound **556** represented by Formula 5.

[0429] In each of the first compound **522** and the fourth compound **552** represented by Formula 1, a difference between a singlet energy level and a triplet energy level can be 0.3 eV or less. As a result, each of the first compound **522** and the fourth compound **552** can have a delayed fluorescent property so that a non-emissive triplet exciton can be converted into a singlet exciton. Accordingly, the emitting efficiency of the OLED D**5** and the organic light emitting display device **100** including the OLED D**5** can be improved.

[0430] Since the first blue EML **520** includes the first compound **522** as an n-type host and the second compound **524** as a p-type host, the exciton generation efficiency in the first blue EML **520** and an exciton transfer efficiency into the third compound **526** can be improved. Since the second blue EML **550** includes the fourth compound **552** as an n-type host and the fifth compound **554** as a p-type host, the exciton generation efficiency in the second blue EML **550** and an exciton transfer efficiency into the sixth compound **556** can be improved. Accordingly, the emitting efficiency of the OLED D**5**, where the first blue EML **520** includes the first, second and third compounds **522**, **524** and **526** and the second blue EML **550** includes the fourth, fifth and sixth compounds **552**, **554** and **556** and the organic light emitting display device **100** including the OLED D**5** can be improved.

[0431] Moreover, since each of the first compound **522** and the fourth compound **552** represented by formula 1 includes a tetraarylsilane structure represented by Formula 1-1, a boron atom can be protected. Accordingly, the lifespan of the OLED D**5** including the first compound **522** and the fourth compound **552** and the organic light emitting display device **100** including the OLED D**5** can be improved.

[0432] Further, since a complex formation between the first compound **522** represented by formula 1 and the third compound **526** represented by formula 5 and a complex formation between the fourth compound **552** represented by formula 1 and the sixth compound **556** represented by formula 5 can be prevented by a tetraarylsilane structure, a concentration of the third compound **526** in the first blue EML **520** and a concentration of the sixth compound **556** in the second blue EML **550** can be increased. Accordingly, the emitting efficiency of the OLED D**5** and the organic light emitting display device **100** including the OLED D**5** can be further improved.

[0433] Furthermore, since each of the first compound **522** and the fourth compound **552** represented by Formula 1 has a LUMO energy level higher than that of the third compound **526** and the sixth compound **556**, respectively, an electron trap in the first compound **522** and the fourth compound **552** can be suppressed. Accordingly, the driving voltage of the OLED D**5** and the organic light emitting display device **100** including the OLED D**5** can be reduced.

[0434] In an aspect of the present disclosure, the first blue EML **520** can include the first compound **522** represented by Formula 1, the second compound **524** represented by Formula 3, the third compound **526** represented by Formula 5 and the first fluorescent compound represented by Formula 7, and the second blue EML **550** can include the fourth compound **552** represented by Formula 1, the fifth compound **554** represented by Formula 3, the sixth compound **556** represented by Formula 5 and the second fluorescent compound represented by Formula 7.

[0435] In each of the first compound **522** and the fourth compound **552** represented by Formula 1 and the first and second fluorescent compounds represented by Formula 7, a difference between a singlet energy level and a triplet energy level can be 0.3 eV or less. As a result, each of the first compound **522** and the fourth compound **552** and the first and second fluorescent compounds can have a delayed fluorescent property so that a non-emissive triplet exciton can be converted into a singlet exciton. Accordingly, the emitting efficiency of the OLED D**5** and the organic light emitting display device **100** including the OLED D**5** can be improved.

[0436] Since the first blue EML **520** includes the first compound **522** as an n-type host and the second compound **524** as a p-type host, the exciton generation efficiency in the first blue EML **520** and an exciton transfer efficiency into the third compound **526** and/or the first fluorescent compound can be improved. In addition, since the second blue EML **550** includes the fourth compound **552** as an n-type host and the fifth compound **554** as a p-type host, the exciton generation efficiency in the second blue EML **550** and an exciton transfer efficiency into the sixth compound **556** and/or the second fluorescent compound can be improved. Accordingly, the emitting efficiency of the OLED D**5** and the organic light emitting display device **100** including the OLED D**5** can be improved.

[0437] Moreover, since each of the first and fourth compounds **522** and **552** represented by formula 1 includes a tetraarylsilane structure represented by Formula 1-1, a boron atom can be protected. Accordingly, the lifespan of the OLED D**5** and the organic light emitting display device **100** including the OLED D**5** can be improved.

[0438] Further, since a complex formation between the first compound **522** represented by formula 1 and the third compound **526** represented by formula 5 and a complex formation between the fourth compound **552** represented by formula 1 and the sixth compound **556** represented by formula 5 can be prevented by a tetraarylsilane structure, a concentration of the third compound **526** in the first blue EML **520** and a concentration of the sixth compound **556** in the second blue EML **550** can be increased. Accordingly, the emitting efficiency of the OLED D**5** and the organic light emitting display device **100** including the OLED D**5** can be further improved.

[0439] Furthermore, since the first and fourth compounds **522** and **552** represented by Formula 1 has a LUMO energy level higher than that of the third and sixth compounds **526** and **556** and the fluorescent compound, respectively, an electron trap in the first and fourth compounds **522** and **552** can be suppressed. Accordingly, the driving voltage of the OLED D**5** and the organic light emitting display device **100** including the OLED D**5** can be reduced.

[0440] FIG. **10** is a schematic cross-sectional view illustrating an organic light emitting display device according to a seventh embodiment of the present disclosure.

[0441] As illustrated in FIG. **10**, the organic light emitting display device **600** includes a first substrate **610**, where a red pixel region RP, a green pixel region GP and a blue pixel region BP are defined, a second substrate **670** facing the first substrate **610**, an OLED D, which is positioned between the first and second substrates **610** and **670** and provides white emission, and a color conversion layer **680** between the OLED D and the second substrate **670**.

[0442] Optionally, a color filter can be formed between the second substrate **670** and each color conversion layer **680**.

[0443] Each of the first and second substrates **610** and **670** can be a glass substrate or a flexible substrate. For example, the flexible substrate can be a polyimide (PI) substrate, a polyethersulfone (PES) substrate, a polyethylenenaphthalate (PEN) substrate, a polyethylene terephthalate (PET) substrate or a polycarbonate (PC) substrate.

[0444] A TFT Tr, which corresponds to each of the red, green and blue pixel regions RP, GP and BP, is formed on the first substrate **610**, and a planarization layer **650**, which has a drain contact hole **652** exposing an electrode, e.g., a drain electrode, of the TFT Tr is formed to cover the TFT Tr.

[0445] The OLED D including a first electrode **210**, an organic light emitting layer **220** and a second electrode **230** is formed on the planarization layer **650**. In this instance, the first electrode **210** can be connected to the drain electrode of the TFT Tr through the drain contact hole **652**.

[0446] The first electrode **210** can be an anode, and the second electrode **230** can be a cathode. One of the first and second electrodes **210** and **230** can be a reflective electrode, and the other one of the first and second electrodes **210** and **230** can be a transparent (or a semi-transparent) electrode. For example, the first electrode **210** can have a single-layered structure of ITO, and the second electrode **230** can be formed of Al.

[0447] A bank layer **666** is formed on the planarization layer **650** to cover an edge of the first electrode **210**. Namely, the bank layer **666** is positioned at a boundary of the pixel region and exposes a center of the first electrode **210** in the pixel region. Since the OLED D emits blue light in each of the red, green and blue pixel regions RP, GP and BP, the organic light emitting layer **220** can be integrally formed as a common layer in the red, green and blue pixel regions RP, GP and BP without separation. The bank layer **666** can be formed to prevent a current leakage at an edge of the first electrode **210** and can be omitted.

[0448] The OLED D emits a blue light and can have a structure shown in FIGS. **3**, **5** and **7** to **9**. Namely, the OLED D is formed in each of the red, green and blue pixel regions RP, GP and BP and provides the blue light.

[0449] For example, referring to FIG. **3**, the organic light emitting layer **220** of the OLED D includes the blue EML **240**A, and the blue EML **240**A include the first compound **242***a* represented by Formula 1, the second compound **244***a* represented by Formula 3 and the third compound **246***a* represented by Formula 5.

[0450] The first compound **242***a* can have a delayed fluorescent property so that the emitting efficiency of the OLED D and the organic light emitting display device **600** including the OLED D can be improved.

[0451] In addition, since the first compound **242***a* represented by formula 1 includes a tetraarylsilane structure represented by Formula 1-1, a boron atom can be protected. Accordingly, the lifespan of the OLED D and the organic light emitting display device **600** including the OLED D can be improved.

[0452] Moreover, since a complex formation between the first compound **242***a* represented by formula 1 and the third compound **246***a* represented by formula 5 can be prevented by a tetraarylsilane structure, a concentration of the third compound **246***a* in the blue EML **240**A can be increased. Accordingly, the emitting efficiency of the OLED D and the organic light emitting display device **600** including the OLED D can be further improved.

[0453] Furthermore, since the first compound **242***a* represented by Formula 1 has a LUMO energy level higher than that of the third compound **246***a*, an electron trap in the first compound **242***a* can be suppressed. Accordingly, the driving voltage of the OLED D and the organic light emitting display device **600** including the OLED D can be reduced.

[0454] In an aspect of the present disclosure, as shown in FIG. **5**, the organic light emitting layer **220** of the OLED D includes the blue EML **240**B, and the blue EML **240**B include the first compound **242***b* represented by Formula 1, the second compound **244***b* represented by Formula 3, the third compound **246***b* represented by Formula 5 and the fluorescent compound **248***b* represented by Formula 7.

[0455] Each of the first compound **242***b* represented by Formula 1 and the fluorescent compound **248***b* represented by Formula 7 can have a delayed fluorescent property so that the emitting efficiency of the OLED D and the organic light emitting display device **600** including the OLED D can be improved.

[0456] In addition, since the first compound **242***b* represented by formula 1 includes a tetraarylsilane structure represented by Formula 1-1, a boron atom can be protected. Accordingly, the lifespan of the OLED D and the organic light emitting display device **600** including the OLED D can be improved.

[0457] Moreover, since a complex formation between the first compound **242***b* represented by formula 1 and the third compound **246***b* represented by formula 5 can be prevented by a tetraarylsilane structure, a concentration of the third compound **246***b* in the EML **240**B can be increased. Accordingly, the emitting efficiency of the OLED D and the organic light emitting display device **600** including the OLED D can be further improved.

[0458] Furthermore, since the first compound **242***b* represented by Formula 1 has a LUMO energy level higher than that of each of the third and fluorescent compounds **246***b* and **248***b*, an electron trap in the first compound **242***b* can be suppressed. Accordingly, the driving voltage of the OLED D and the organic light emitting display device **600** including the OLED D can be reduced.

[0459] The color conversion layer **680** includes a first color conversion layer **682** corresponding to the red pixel region RP and a second color conversion layer **684** corresponding to the green pixel region GP. For example, the color conversion layer **680** can include an inorganic color conversion material such as a quantum dot. The color conversion layer is not presented in the blue pixel region BP so that the OLED D in the blue pixel region BP can directly face the second substrate **670**.

[0460] The blue light from the OLED D is converted into the red light by the first color conversion layer **682** in the red pixel region RP, and the blue light from the OLED D is converted into the green light by the second color conversion layer **684** in the green pixel region GP.

[0461] Accordingly, the organic light emitting display device **600** can display a full-color image.

[0462] When the light from the OLED D passes through the first substrate **610** to display an image, the color conversion layer **680** can be disposed between the OLED D and the first substrate **610**.

[0463] FIG. **11** is a schematic cross-sectional view illustrating an organic light emitting display device according to an eighth embodiment of the present disclosure.

[0464] As illustrated in FIG. **11**, the organic light emitting display device **700** includes a first substrate **710**, where a red pixel region RP, a green pixel region GP and a blue pixel region BP are defined, a second substrate **770** facing the first substrate **710**, an OLED D, which is positioned between the first and second substrates **710** and **770** and provides white emission, and a color filter layer **780** between the OLED D and the second substrate **770**.

[0465] Each of the first and second substrates **710** and **770** can be a glass substrate or a flexible substrate. For example, the flexible substrate can be a polyimide (PI) substrate, a polyethersulfone (PES) substrate, a polyethylenenaphthalate (PEN) substrate, a polyethylene terephthalate (PET) substrate or a polycarbonate (PC) substrate.

[0466] A buffer layer **720** is formed on the first substrate **710**, and the TFT Tr corresponding to each of the red, green and blue pixel regions RP, GP and BP is formed on the buffer layer **720**. The buffer layer **720** can be omitted.

[0467] A semiconductor layer **722** is formed on the buffer layer **720**. The semiconductor layer **722** can include an oxide semiconductor material or polycrystalline silicon.

[0468] A gate insulating layer **724** is formed on the semiconductor layer **722**. The gate insulating layer **724** can be formed of an inorganic insulating material such as silicon oxide or silicon nitride.

[0469] A gate electrode **730**, which is formed of a conductive material, e.g., metal, is formed on the gate insulating layer **724** to correspond to a center of the semiconductor layer **722**.

[0470] An interlayer insulating layer **732**, which is formed of an insulating material, is formed on the gate electrode **730**. The interlayer insulating layer **732** can be formed of an inorganic insulating material, e.g., silicon oxide or silicon nitride, or an organic insulating material, e.g., benzocyclobutene or photo-acryl.

[0471] The interlayer insulating layer **732** includes first and second contact holes **734** and **736** exposing both sides of the semiconductor layer **722**. The first and second contact holes **734** and **736** are positioned at both sides of the gate electrode **730** to be spaced apart from the gate electrode **730**.

[0472] A source electrode **740** and a drain electrode **742**, which are formed of a conductive material, e.g., metal, are formed on the interlayer insulating layer **732**.

[0473] The source electrode **740** and the drain electrode **742** are spaced apart from each other with respect to the gate electrode **730** and respectively contact both sides of the semiconductor layer **722** through the first and second contact holes **734** and **736**.

[0474] The semiconductor layer **722**, the gate electrode **730**, the source electrode **740** and the drain electrode **742** constitute the TFT Tr. The TFT Tr serves as a driving element. Namely, the TFT Tr can correspond to the driving TFT Td (of FIG. **1**).

[0475] Optionally, the gate line and the data line cross each other to define the pixel, and the switching TFT is formed to be connected to the gate and data lines. The switching TFT is connected to the TFT Tr as the driving element.

[0476] In addition, the power line, which can be formed to be parallel to and spaced apart from one of the gate and data lines, and the storage capacitor for maintaining the voltage of the gate electrode of the TFT Tr in one frame can be further formed.

[0477] A planarization layer **750**, which includes a drain contact hole **752** exposing the drain electrode **742** of the TFT Tr, is formed to cover the TFT Tr.
[0478] A first electrode **810**, which is connected to the drain electrode **742** of the TFT Tr through the drain contact hole **752**, is separately formed in each pixel region and on the planarization layer **750**. The first electrode **810** can be an anode and can be formed of a conductive material having a relatively high work function. For example, the first electrode **810** can include a transparent conductive oxide layer formed of a transparent conductive oxide (TCO).
[0479] For example, the transparent conductive oxide material layer of the first electrode **810** can be formed of one of indium-tin-oxide (ITO), indium-zinc-oxide (IZO), indium-tin-zinc-oxide (ITZO), tin oxide (SnO), zinc oxide (ZnO), indium-copper-oxide (ICO) and aluminum-zinc-oxide (Al:ZnO, AZO).
[0480] The first electrode **810** can further include a reflective layer to have a double-layered structure or a triple-layered structure. Namely, the first electrode **810** can be a reflective electrode.
[0481] For example, the reflective layer can be formed of one of silver (Ag), an alloy of Ag and one of palladium (Pd), copper (Cu), indium (In) and neodymium (Nd), and aluminum-palladium-copper (APC) alloy. For example, the first electrode **810** can have a double-layered structure of Ag/ITO or APC/ITO or a triple-layered structure of ITO/Ag/ITO or ITO/APC/ITO.
[0482] A bank layer **766** is formed on the planarization layer **750** to cover an edge of the first electrode **810**. Namely, the bank layer **766** is positioned at a boundary of the pixel region and exposes a center of the first electrode **810** in the pixel region. Since the OLED D emits blue light in each of the red, green and blue pixel regions RP, GP and BP, an organic light emitting layer **820** can be integrally formed as a common layer in the red, green and blue pixel regions RP, GP and BP without separation. The bank layer **766** can be formed to prevent a current leakage at an edge of the first electrode **810** and can be omitted.
[0483] The organic light emitting layer **820** is formed on the first electrode **810**.
[0484] A second electrode **830** is formed over the first substrate **710** where the organic light emitting layer **820** is formed.
[0485] In the organic light emitting display device **700**, since the light emitted from the organic light emitting layer **820** is incident to the color filter layer **780** through the second electrode **830**, the second electrode **830** has a thin profile for transmitting the light.
[0486] The first electrode **810**, the organic light emitting layer **820** and the second electrode **830** constitute the OLED D.
[0487] The color filter layer **780** is disposed over the OLED D and includes a red color filter **782**, a green color filter **784** and a blue color filter **786** respectively corresponding to the red pixel region RP, the green pixel region GP and the blue pixel region BP. The red color filter **782** includes at least one of a red dye and a red pigment, the green color filter **784** includes at least one of a green dye and a green pigment, and the blue color filter **786** includes at least one of a blue dye and a blue pigment.
[0488] The color filter layer **780** can be attached to the OLED D using an adhesive layer. Alternatively, the color filter layer **780** can be formed directly on the OLED D.
[0489] An encapsulation layer (or an encapsulation film) can be formed to prevent penetration of moisture into the OLED D. For example, the encapsulation layer can include a first inorganic insulating layer, an organic insulating layer and a second inorganic insulating layer sequentially stacked, but it is not limited thereto. The encapsulation layer can be omitted.
[0490] In the bottom-emission type organic light emitting display device **700**, a metal plate can be disposed over the second electrode **830**. The metal plate can be attached to the OLED D using an adhesive layer.
[0491] A polarization plate for reducing an ambient light reflection can be disposed over the top-emission type OLED D. For example, the polarization plate can be a circular polarization plate.
[0492] In the OLED D of FIG. **11**, the first electrode **810** and the second electrode **830** are a

reflective electrode and a transparent (or semitransparent) electrode, respectively, and the color filter layer **780** is disposed over the OLED D.

[0493] Alternatively, the first electrode **810** and the second electrode **830** can be a transparent (or semitransparent) electrode and a reflective electrode, respectively, and the color filter layer **780** can be disposed between the OLED D and the first substrate **710**. In this case, first electrode **810** can have a single-layered structure of the transparent conductive oxide layer.

[0494] A color conversion layer can be formed between the OLED D and the color filter layer **780**. The color conversion layer can include a red color conversion layer, a green color conversion layer and a blue color conversion layer respectively corresponding to the red, green and blue pixel regions RP, GP and BP. The white light from the OLED D is converted into the red light, the green light and the blue light by the red, green and blue color conversion layer, respectively.

[0495] The color conversion layer can be included instead of the color filter layer **780**.

[0496] As described above, in the organic light emitting display device **700**, the OLED D in the red, green and blue pixel regions RP, GP and BP emits the white light, and the white light from the OLED D passes through the red color filter **782**, the green color filter **784** and the blue color filter **786**. As a result, the red light, the green light and the blue light are provided from the red pixel region RP, the green pixel region GP and the blue pixel region BP, respectively.

[0497] In FIG. **11**, the OLED D emitting the white light is used for a display device. Alternatively, the OLED D can be formed on an entire surface of a substrate without at least one of the driving element and the color filter layer to be used for a lightening device. The display device and the lightening device each including the OLED D of the present disclosure can be referred to as an organic light emitting device.

[0498] FIG. **12** is a schematic cross-sectional view of an OLED according to a ninth embodiment of the present disclosure.

[0499] As illustrated in FIG. **12**, the OLED D**6** includes first and second electrodes **810** and **830**, which face each other, and an organic light emitting layer **820** therebetween. The organic light emitting layer **820** includes a first emitting part **910** including a first blue EML **920**, e.g., a first blue EML, a second emitting part **930** including a second blue EML **940**, e.g., a second blue EML, and a third emitting part **950** including a third EML **960**. The organic light emitting layer **820** can further includes a first CGL **970** between the first and third emitting parts **910** and **950** and a second CGL **980** between the second and third emitting parts **930** and **950**. The OLED D**6** can further include a capping layer on the second electrode **830** to enhance a light extraction efficiency.

[0500] The organic light emitting display device **700** can include a red pixel region, a green pixel region and a blue pixel region, and the OLED D**6** can be positioned in the red, green and blue pixel regions RP, GP and BP and emits blue light.

[0501] The first electrode **810** can be an anode, and the second electrode **830** can be a cathode. One of the first and second electrodes **810** and **830** can be a reflective electrode, and the other one of the first and second electrodes **810** and **830** can be a transparent (or a semi-transparent) electrode. For example, the first electrode **810** can have a single-layered structure of ITO, and the second electrode **830** can be formed of Al.

[0502] The first emitting part **910** can further include at least one of a first HTL **914** under the first blue EML **920** and a first ETL **916** over the first blue EML **920**.

[0503] In addition, the first emitting part **910** can further include an HIL **912** between the first electrode **810** and the first HTL **914**.

[0504] Moreover, the first emitting part **910** can further include at least one of a first EBL between the first HTL **914** and the first blue EML **920** and a first HBL between the first blue EML **920** and the first ETL **916**.

[0505] The second emitting part **930** can further include at least one of a second HTL **932** under the second blue EML **940** and a second ETL **934** over the second blue EML **940**.

[0506] In addition, the second emitting part **930** can further include an EIL **936** between the second electrode **830** and the second ETL **934**.

[0507] Moreover, the second emitting part **930** can further include at least one of a second EBL between the second HTL **932** and the second blue EML **940** and a second HBL between the second blue EML **940** and the second ETL **934**.

[0508] In the third emitting part **950**, the third EML **960** can include a red EML **962**, a yellow-green EML **964** and a green EML **966**. In this case, the yellow-green EML **964** is disposed between the red and green EMLs **962** and **966**. Alternatively, the yellow-green EML **964** can be omitted, and the third EML **960** can have a double-layered structure including the red and green EMLs **962** and **966**.

[0509] The red EML **962** includes a red host and a red dopant, the green EML **966** includes a green host and a green dopant, and the yellow-green EML **964** includes a yellow-green host and a yellow-green dopant. Each of the red dopant, the green dopant and the yellow-green dopant can be one of a fluorescent compound, a phosphorescent compound and a delayed fluorescent compound.

[0510] For example, the red host can be selected from the group consisting of mCP-CN, CBP, mCBP, mCP, DPEPO, 2,8-bis(diphenylphosphoryl)dibenzothiophene (PPT), 1,3,5-tri[(3-pyridyl)-phen-3-yl]benzene (TmPyPB), 2,6-di(9H-carbazol-9-yl)pyridine (PYD-2Cz), 2,8-di(9H-carbazol-9-yl)dibenzothiophene (DCzDBT), 3′,5′-di(carbazol-9-yl)-[1,1′-biphenyl]-3,5-dicarbonitrile (DCzTPA), 4′-(9H-carbazol-9-yl)biphenyl-3,5-dicarbonitrile(4′-(9H-carbazol-9-yl)biphenyl-3,5-dicarbonitrile (pCzB-2CN), 3′-(9H-carbazol-9-yl)biphenyl-3,5-dicarbonitrile (mCzB-2CN), TSPO1, 9-(9-phenyl-9H-carbazol-6-yl)-9H-carbazole (CCP), 4-(3-(triphenylen-2-yl)phenyl)dibenzo[b,d]thiophene, 9-(4-(9H-carbazol-9-yl)phenyl)-9H-3,9′-bicarbazole, 9-(3-(9H-carbazol-9-yl)phenyl)-9H-3,9′-bicarbazole, 9-(6-(9H-carbazol-9-yl)pyridin-3-yl)-9H-3,9′-bicabazole, 9,9′-diphenyl-9H,9′H-3,3′-bicarbazole (BCzPh), 1,3,5-tris(carbazole-9-yl)benzene (TCP), TCTA, 4,4′-bis(carbazole-9-yl)-2,2′-dimethylbiphenyl (CDBP), 2,7-bis(carbazole-9-yl)-9,9-dimethylfluorene (DMFL-CBP), 2,2′,7,7′-tetrakis(carbazole-9-yl)-9,9-spiorofluorene (Spiro-CBP), and 3,6-bis(carbazole-9-yl)-9-(2-ethyl-hexyl)-9H-carbazole (TCzl), but it is not limited thereto.

[0511] The red dopant can be selected from the group consisting of [bis(2-(4,6-dimethyl)phenylquinoline)](2,2,6,6-tetramethylheptane-3,5-dionate)iridium(III), bis[2-(4-n-hexylphenyl)quinoline](acetylacetonate)iridium(III) (Hex-Ir(phq)2(acac)), tris[2-(4-n-hexylphenyl)quinoline]iridium(III) (Hex-Ir(phq)3), tris[2-phenyl-4-methylquinoline]iridium(III) (lr(Mphq)3), bis(2-phenylquinoline)(2,2,6,6-tetramethylheptene-3,5-dionate)iridium(III) (lr(dpm)PQ2), bis(phenylisoquinoline)(2,2,6,6-tetramethylheptene-3,5-dionate)iridium(III) (Ir(dpm)(piq)2), bis[(4-n-hexylphenyl)isoquinoline](acetylacetonate)iridium(III) (Hex-Ir(piq)2(acac)), tris[2-(4-n-hexylphenyl)quinoline]iridium(III) (Hex-Ir(piq)3), tris(2-(3-methylphenyl)-7-methyl-quinolato)iridium (lr(dmpq)3), bis[2-(2-methylphenyl)-7-methyl-quinoline](acetylacetonate)iridium(III) (Ir(dmpq)2(acac)), bis[2-(3,5-dimethylphenyl)-4-methyl-quinoline](acetylacetonate)iridium(III) (Ir(mphmq)2(acac)), and tris(dibenzoylmethane)mono(1,10-phenanthroline)europium(III) (Eu(dbm)3(phen)), but it is not limited thereto.

[0512] Each of the green host and the yellow-green host can be independently selected from the group consisting of mCP-CN, CBP, mCBP, mCP, DPEPO, 2,8-bis(diphenylphosphoryl)dibenzothiophene (PPT), TmPyPB, PYD-2Cz, 2,8-di(9H-carbazol-9-yl)dibenzothiophene (DCzDBT), 3′,5′-di(carbazol-9-yl)-[1,1′-biphenyl]-3,5-dicarbonitrile (DCzTPA), 4′-(9H-carbazol-9-yl)biphenyl-3,5-dicarbonitrile(4′-(9H-carbazol-9-yl)biphenyl-3,5-dicarbonitrile (pCzB-2CN), 3′-(9H-carbazol-9-yl)biphenyl-3,5-dicarbonitrile (mCzB-2CN), TSPO1, and 9-(9-phenyl-9H-carbazol-6-yl)-9H-carbazole (CCP), but it is not limited thereto.

[0513] The green dopant can be selected from the group consisting of [bis(2-phenylpyridine)](pyridyl-2-benzofuro[2,3-b]pyridine)iridium, tris[2-phenylpyridine]iridium(III) (Ir(ppy)3), fac-tris(2-phenylpyridine)iridium(III) (fac-Ir(ppy)3), bis(2-phenylpyridine)(acetylacetonate)iridium(III) (Ir(ppy)2(acac)), tris[2-(p-tolyl)pyridine]iridium(III) (Ir(mppy)3), bis(2-(naphthalene-2-

yl)pyridine)(acetylacetonate)iridium(III) (Ir(npy)2acac), tris(2-phenyl-3-methyl-pyridine)iridium (Ir(3mppy)3), and fac-Tris(2-(3-p-xylyl)phenyl)pyridine iridium(III) (TEG), but it is not limited thereto.
[0514] The yellow-green dopant can be selected from the group consisting of 5,6,11,12-tetraphenylnaphthalene (Rubrene), 2,8-di-tert-butyl-5,11-bis(4-tert-butylphenyl)-6,12-diphenyltetracene (TBRb), bis(2-phenylbenzothiazolato)(acetylacetonate)irdium(III) (Ir(BT)2(acac)), bis(2-(9,9-diethytl-fluoren-2-yl)-1-phenyl-1H-benzo[d]imdiazolato)(acetylacetonate)iridium(III) (Ir(fbi)2(acac)), bis(2-phenylpyridine)(3-(pyridine-2-yl)-2H-chromen-2-onate)iridium(III) (fac-Ir(ppy)2Pc), bis(2-(2,4-difluorophenyl)quinoline)(picolinate)iridium(III) (FPQIrpic), and bis(4-phenylthieno[3,2-c]pyridinato-N,C2′) (acetylacetonate) iridium(III) (PO-01), but it is not limited thereto.
[0515] The third emitting part **950** can include at least one of a third HTL **952** under the third EML **960** and a third ETL **954** over the third EML **960**.
[0516] In addition, the third emitting part **950** can further include at least one of a third EBL between the third HTL **952** and the third EML **960** and a third HBL between the third EML **960** and the third ETL **954**.
[0517] For example, the HIL **912** can include the above hole injection material and can have a thickness of 1 nm to 20 nm, e.g., 5 nm to 15 nm.
[0518] Each of the first to third HTLs **914**, **932** and **952** can include the above hole transporting material and can have a thickness of 30 nm to 60 nm, e.g., 40 nm to 50 nm.
[0519] Each of the first to third ETLs **916**, **934** and **954** can include the above electron transporting material and can have a thickness of 10 nm to 50 nm, e.g., 20 nm to 40 nm.
[0520] The EIL **936** can include the above electron injection material and can have a thickness of 0.1 nm to 10 nm, e.g., 1 nm to 5 nm.
[0521] Each of the first to third EBLs can include the above electron blocking material and can have a thickness of 5 nm to 20 nm, e.g., 5 nm to 15 nm.
[0522] Each of the first to third HBLs can include the above hole blocking material and can have a thickness of 5 nm to 20 nm, e.g., 5 nm to 15 nm.
[0523] The first CGL **970** is positioned between the first and third emitting parts **910** and **950**, and the second CGL **980** is positioned between the second and third emitting parts **930** and **950**. Namely, the first and third emitting parts **910** and **950** can be connected to each other through the first CGL **970**, and the second and third emitting parts **930** and **950** can be connected to each other through the second CGL **980**. The first CGL **970** can be a P-N junction CGL of a first N-type CGL **972** and a first P-type CGL **974**, and the second CGL **980** can be a P-N junction CGL of a second N-type CGL **982** and a second P-type CGL **984**.
[0524] In the first CGL **970**, the first N-type CGL **972** is positioned between the first ETL **916** and the third HTL **952**, and the first P-type CGL **974** is positioned between the first N-type CGL **972** and the third HTL **952**.
[0525] In the second CGL **980**, the second N-type CGL **982** is positioned between the third ETL **954** and the second HTL **932**, and the second P-type CGL **984** is positioned between the second N-type CGL **982** and the second HTL **932**.
[0526] Each of the first and second N-type CGLs **972** and **982** can be an organic layer doped with an alkali metal, e.g., Li, Na, K and Cs, and/or an alkali earth metal, e.g., Mg, Sr, Ba and Ra. For example, each of the first and second N-type CGLs **972** and **982** can be formed of an N-type charge generation material including a host being the organic material, e.g., 4,7-dipheny-1,10-phenanthroline (Bphen) and MTDATA, and a dopant being an alkali metal and/or an alkali earth metal, and the dopant can be doped with a weight % of 0.01 to 30.
[0527] Each of the first and second P-type CGLs **974** and **984** can be formed of a P-type charge generation material including an inorganic material, e.g., tungsten oxide (WO.sub.x), molybdenum oxide (MoO.sub.x), beryllium oxide (Be.sub.2O.sub.3) or vanadium oxide (V.sub.2O.sub.5), an

organic material, e.g., NPD, HAT-CN, F4TCNQ, TPD, TNB, TCTA, N,N′-dioctyl-3,4,9,10-perylenedicarboximide (PTCDI-C8) or their combination.

[0528] The first blue EML **920** can have a thickness in a range of 10 nm to 100 nm, e.g., 10 nm to 50 nm, preferably 20 nm to 40 nm.

[0529] The first blue EML **920** can include a first compound **922**, a second compound **924** and a third compound **926**. The first compound **922** can act as a first host, e.g., an n-type host, the second compound **924** can act as a second host, e.g., a p-type host, and the third compound **926** can act as a dopant.

[0530] The first compound **922** is represented by Formula 1 and can be one of the compounds in Formula 2. The second compound **924** is represented by Formula 3 and can be one of the compounds in Formula 4. The third compound **926** is represented by Formula 5 and can be one of the compounds in Formula 6.

[0531] In the first blue EML **920**, a weight % of each of the first and second compounds **922** and **924** can be greater than that of the third compound **926**. A weight % of the first compound **922** and a weight % of the second compound **924** can be same or different.

[0532] The first compound **922** can have a weight % of 25 to 50, the second compound **924** can have a weight % of 25 to 50, and the third compound **926** can have a weight % of 5 to 25. In an aspect of the present disclosure, a weight % of the first compound **922** and a weight % of the second compound **924** can be same, and the third compound **926** can have a weight % of 8, 12, 16 or 20.

[0533] In an aspect of the present disclosure, the first blue EML **920** can include a first compound **922**, a second compound **924**, a third compound **926** and a first fluorescent compound. The first compound **922** can act as a first host, e.g., an n-type host, and the second compound **924** can act as a second host, e.g., a p-type host. The third compound **926** can act as an auxiliary host or an auxiliary dopant, and the first fluorescent compound can act as a dopant.

[0534] The first compound **922** is represented by Formula 1 and can be one of the compounds in Formula 2. The second compound **924** is represented by Formula 3 and can be one of the compounds in Formula 4. The third compound **926** is represented by Formula 5 and can be one of the compounds in Formula 6. The first fluorescent compound is represented by Formula 7 and can be one of the compounds in Formula 8.

[0535] In the first blue EML **920**, a weight % of each of the first and second compounds **922** and **924** can be greater than that of each of the third compound **926** and the first fluorescent compound. In addition, a weight % of the third compound **926** can be greater than that of the first fluorescent compound. A weight % of the first compound **922** and a weight % of the second compound **924** can be same or different.

[0536] The first compound **922** can have a weight % of 25 to 50, and the second compound **924** can have a weight % of 25 to 50. The third compound **926** can have a weight % of 5 to 25, and the first fluorescent compound can have a weight % of 0.1 to 3. In an aspect of the present disclosure, a weight % of the first compound **922** and a weight % of the second compound **924** can be same, the third compound **926** can have a weight % of 8, 12, 16 or 20, and the first fluorescent compound can have a weight % of 1.

[0537] The second blue EML **940** can have a thickness in a range of 10 nm to 100 nm, e.g., 10 nm to 50 nm, preferably 20 nm to 40 nm.

[0538] The second blue EML **940** can include a fourth compound **942**, a fifth compound **944** and a sixth compound **946**. The fourth compound **942** can act as a first host, e.g., an n-type host, the fifth compound **944** can act as a second host, e.g., a p-type host, and the sixth compound **946** can act as a dopant.

[0539] The fourth compound **942** is represented by Formula 1 and can be one of the compounds in Formula 2. The fifth compound **944** is represented by Formula 3 and can be one of the compounds in Formula 4. The sixth compound **946** is represented by Formula 5 and can be one of the compounds in Formula 6.

[0540] In the second blue EML **940**, a weight % of each of the fourth and fifth compounds **942** and **944** can be greater than that of the sixth compound **946**. A weight % of the fourth compound **942** and a weight % of the fifth compound **944** can be same or different.

[0541] The fourth compound **942** can have a weight % of 25 to 50, the fifth compound **944** can have a weight % of 25 to 50, and the sixth compound **946** can have a weight % of 5 to 25. In an aspect of the present disclosure, a weight % of the fourth compound **942** and a weight % of the fifth compound **944** can be same, and the sixth compound **946** can have a weight % of 8, 12, 16 or 20.

[0542] In an aspect of the present disclosure, the second blue EML **940** can include a fourth compound **942**, a fifth compound **944**, a sixth compound **946** and a second fluorescent compound. The fourth compound **942** can act as a first host, e.g., an n-type host, and the fifth compound **944** can act as a second host, e.g., a p-type host. The sixth compound **946** can act as an auxiliary host or an auxiliary dopant, and the second fluorescent compound can act as a dopant.

[0543] The fourth compound **942** is represented by Formula 1 and can be one of the compounds in Formula 2. The fifth compound **944** is represented by Formula 3 and can be one of the compounds in Formula 4. The sixth compound **946** is represented by Formula 5 and can be one of the compounds in Formula 6. The second fluorescent compound is represented by Formula 7 and can be one of the compounds in Formula 8.

[0544] In the second blue EML **940**, a weight % of each of the fourth and fifth compounds **942** and **944** can be greater than that of each of the sixth compound **946** and the second fluorescent compound. In addition, a weight % of the sixth compound **946** can be greater than that of the second fluorescent compound. A weight % of the fourth compound **942** and a weight % of the fifth compound **944** can be same or different.

[0545] The fourth compound **942** can have a weight % of 25 to 50, and the fifth compound **944** can have a weight % of 25 to 50. The sixth compound **946** can have a weight % of 5 to 25, and the second fluorescent compound can have a weight % of 0.1 to 3. In an aspect of the present disclosure, a weight % of the fourth compound **942** and a weight % of the fifth compound **944** can be same, the sixth compound **946** can have a weight % of 8, 12, 16 or 20, and the second fluorescent compound can have a weight % of 1.

[0546] The first compound **922** in the first blue EML **920** and the fourth compound **942** in the second blue EML **940** can be same or different. The second compound **924** in the first blue EML **920** and the fifth compound **944** in the second blue EML **940** can be same or different. The third compound **926** in the first blue EML **920** and the sixth compound **946** in the second blue EML **940** can be same or different. The first fluorescent compound in the first blue EML **920** and the second fluorescent compound in the second blue EML **940** can be same or different.

[0547] A weight % of the first compound **922** in the first blue EML **920** and a weight % of the fourth compound **942** in the second blue EML **940** can be same or different. A weight % of the second compound **924** in the first blue EML **920** and a weight % of the fifth compound **944** in the second blue EML **940** can be same or different. A weight % of the third compound **926** in the first blue EML **920** and a weight % of the sixth compound **946** in the second blue EML **940** can be same or different. A weight % of the first fluorescent compound in the first blue EML **920** and a weight % of the second fluorescent compound in the second blue EML **940** can be same or different.

[0548] In FIG. **12**, the first blue EML **920** includes the first compound **922** represented by Formula 1, the second compound **924** represented by Formula 3 and the third compound **926** represented by Formula 5 or includes the first compound **922** represented by Formula 1, the second compound **924** represented by Formula 3, the third compound **926** represented by Formula 5 and the first fluorescent compound represented by Formula 7, and the second blue EML **940** includes the fourth compound **942** represented by Formula 1, the fifth compound **944** represented by Formula 3 and the sixth compound **946** represented by Formula 5 or includes the fourth compound **942** represented by Formula 1, the fifth compound **944** represented by Formula 3, the sixth compound

946 represented by Formula 5 and the second fluorescent compound represented by Formula 7.

[0549] Alternatively, one of the first and second blue EMLs **920** and **940** can include the above blue host and the above blue dopant.

[0550] The organic light emitting layer **820** of the OLED D**6** includes the first emitting part **910** including the first blue EML **920**, the second emitting part **930** including the second blue EML **940** and the third emitting part **950** including the red, yellow-green and green EMLs **962**, **964** and **966** so that the OLED D**6** has a tandem structure.

[0551] In this case, the first blue EML **920** can include the first compound **922** represented by Formula 1, the second compound **924** represented by Formula 3 and the third compound **926** represented by Formula 5, and the second blue EML **940** can include the fourth compound **942** represented by Formula 1, the fifth compound **944** represented by Formula 3 and the sixth compound **946** represented by Formula 5.

[0552] In each of the first compound **922** and the fourth compound **942** represented by Formula 1, a difference between a singlet energy level and a triplet energy level can be 0.3 eV or less. As a result, each of the first compound **922** and the fourth compound **942** can have a delayed fluorescent property so that a non-emissive triplet exciton can be converted into a singlet exciton. Accordingly, the emitting efficiency of the OLED D**6** and the organic light emitting display device **700** including the OLED D**6** can be improved.

[0553] Since the first blue EML **920** includes the first compound **922** as an n-type host and the second compound **924** as a p-type host, the exciton generation efficiency in the first blue EML **920** and an exciton transfer efficiency into the third compound **926** can be improved. Since the second blue EML **940** includes the fourth compound **942** as an n-type host and the fifth compound **944** as a p-type host, the exciton generation efficiency in the second blue EML **940** and an exciton transfer efficiency into the sixth compound **946** can be improved. Accordingly, the emitting efficiency of the OLED D**6**, where the first blue EML **920** includes the first, second and third compounds **922**, **924** and **926** and the second blue EML includes the fourth, fifth and sixth compounds **942**, **944** and **946** and the organic light emitting display device **700** including the OLED D**6** can be improved.

[0554] Moreover, since each of the first compound **922** and the fourth compound **942** represented by formula 1 includes a tetraarylsilane structure represented by Formula 1-1, a boron atom can be protected. Accordingly, the lifespan of the OLED D**6** including the first compound **922** and the fourth compound **942** and the organic light emitting display device **700** including the OLED D**6** can be improved.

[0555] Further, since a complex formation between the first compound **922** represented by formula 1 and the third compound **926** represented by formula 5 and a complex formation between the fourth compound **942** represented by formula 1 and the sixth compound **946** represented by formula 5 can be prevented by a tetraarylsilane structure, a concentration of the third compound **926** in the first blue EML **920** and a concentration of the sixth compound **946** in the second blue EML **940** can be increased. Accordingly, the emitting efficiency of the OLED D**6** and the organic light emitting display device **700** including the OLED D**6** can be further improved.

[0556] Furthermore, since the first compound **922** and the fourth compound **942** represented by Formula 1 has a LUMO energy level higher than the third compound **926** and the sixth compound **946**, respectively, an electron trap in the first compound **922** and the fourth compound **942** can be suppressed. Accordingly, the driving voltage of the OLED D**6** and the organic light emitting display device **700** including the OLED D**6** can be reduced.

[0557] In an aspect of the present disclosure, the first blue EML **920** can include the first compound **922** represented by Formula 1, the second compound **924** represented by Formula 3, the third compound **926** represented by Formula 5 and the first fluorescent compound represented by Formula 7, and the second blue EML **940** can include the fourth compound **942** represented by Formula 1, the fifth compound **944** represented by Formula 3, the sixth compound **946** represented by Formula 5 and the second fluorescent compound represented by Formula 7.

[0558] In each of the first compound **922** and the fourth compound **942** represented by Formula 1 and the first and second fluorescent compounds represented by Formula 7, a difference between a singlet energy level and a triplet energy level can be 0.3 eV or less. As a result, each of the first compound **922** and the fourth compound **942** and the first and second fluorescent compounds can have a delayed fluorescent property so that a non-emissive triplet exciton can be converted into a singlet exciton. Accordingly, the emitting efficiency of the OLED D**6** and the organic light emitting display device **700** including the OLED D**6** can be improved.
[0559] Since the first blue EML **920** includes the first compound **922** as an n-type host and the second compound **924** as a p-type host, the exciton generation efficiency in the first blue EML **920** and an exciton transfer efficiency into the third compound **926** and/or the first fluorescent compound can be improved. In addition, since the second blue EML **940** includes the fourth compound **942** as an n-type host and the fifth compound **944** as a p-type host, the exciton generation efficiency in the second blue EML **940** and an exciton transfer efficiency into the sixth compound **946** and/or the second fluorescent compound can be improved. Accordingly, the emitting efficiency of the OLED D**6** and the organic light emitting display device **700** including the OLED D**6** can be improved.
[0560] Moreover, since each of the first and fourth compounds **922** and **942** represented by formula 1 includes a tetraarylsilane structure represented by Formula 1-1, a boron atom can be protected. Accordingly, the lifespan of the OLED D**6** and the organic light emitting display device **700** including the OLED D**6** can be improved.
[0561] Further, since a complex formation between the first compound **922** represented by formula 1 and the third compound **926** represented by formula 5 and a complex formation between the fourth compound **942** represented by formula 1 and the sixth compound **946** represented by formula 5 can be prevented by a tetraarylsilane structure, a concentration of the third compound **926** in the first blue EML **920** and a concentration of the sixth compound **946** in the second blue EML **940** can be increased. Accordingly, the emitting efficiency of the OLED D**6** and the organic light emitting display device **700** including the OLED D**6** can be further improved.
[0562] Furthermore, since the first and fourth compounds **922** and **942** represented by Formula 1 has a LUMO energy level higher than that of each of the third and sixth compounds **926** and **946** and the first and second fluorescent compounds, respectively, an electron trap in the first and fourth compounds **922** and **942** can be suppressed. Accordingly, the driving voltage of the OLED D**6** and the organic light emitting display device **700** including the OLED D**6** can be reduced.
[0563] It will be apparent to those skilled in the art that various modifications and variations can be made in the present disclosure without departing from the spirit or scope of the present disclosure. Thus, it is intended that the present disclosure cover the modifications and variations of this disclosure provided they come within the scope of the appended claims and their equivalents.

Claims

1. An organic light emitting diode, comprising: a first electrode; a second electrode facing the first electrode; and a first blue emitting material layer including a first compound, a second compound and a third compound, and positioned between the first and second electrodes, wherein the first compound is represented by Formula 1: ##STR00047## wherein in Formula 1, at least one of R.sub.1 to R.sub.11 is represented by Formula 1-1, at least another one of R.sub.1 to R.sub.11 is represented by Formula 1-2, each of the rest of R.sub.1 to R.sub.11 is independently selected from the group consisting of hydrogen, deuterium, halogen, cyano, a substituted or unsubstituted C1 to C20 alkyl group, a substituted or unsubstituted C3 to C30 cycloalkyl group, a substituted or unsubstituted C1 to C20 alkoxy group, a substituted or unsubstituted C1 to C20 alkylamino group, a substituted or unsubstituted C6 to C30 arylamino group, a substituted or unsubstituted C1 to C20 alkylsilyl group, a substituted or unsubstituted C6 to C30 arylsilyl group, a substituted or unsubstituted C6 to C30 aryl group and a substituted or unsubstituted C3 to C30 heteroaryl group, or optionally, adjacent two of the rest of R.sub.1 to R.sub.11 are combined to form a ring, each of Y.sub.1 and Y.sub.2 is independently selected from O or NR.sub.12, R.sub.12 is selected from the group consisting of a substituted or unsubstituted C1 to C20 alkyl group, a substituted or unsubstituted C3 to C30 cycloalkyl group, a substituted or unsubstituted C1 to C20 alkoxy group, a substituted or unsubstituted C1 to C20 alkylamino group, a substituted or unsubstituted C6 to C30 arylamino group, a substituted or unsubstituted C1 to C20 alkylsilyl group, a substituted or unsubstituted C6 to C30 arylsilyl group, a substituted or unsubstituted C6 to C30 aryl group and a substituted or unsubstituted C3 to C30 heteroaryl group, ##STR00048## wherein in Formula 1-1, L is selected from a substituted or unsubstituted C6 to C30 arylene group, and each of Ar.sub.1, Ar.sub.2 and Ar.sub.3 is independently selected from a substituted or unsubstituted C6 to C30 aryl group, wherein in Formula 1-2, each of a1 and a4 is independently an integer of 0 to 4, each of a2 and a3 is independently an integer of 0 to 2, when a1 is 2 or more, two or more R.sub.21 are same or different, when a2 is 2, two R.sub.22 are same or different, when a3 is 2, two R.sub.23 are same or different, when a4 is 2 or more, two or more R.sub.24 are same or different, each of R.sub.21, R.sub.22, R.sub.23 and R.sub.24 is independently selected from the group consisting of deuterium, halogen, cyano, a substituted or unsubstituted C1 to C20 alkyl group, a substituted or unsubstituted C3 to C30 cycloalkyl group, a substituted or unsubstituted C1 to C20 alkoxy group, a substituted or unsubstituted C1 to C20 alkylamino group, a substituted or unsubstituted C6 to C30 arylamino group, a substituted or unsubstituted C1 to C20 alkylsilyl group, a substituted or unsubstituted C6 to C30 arylsilyl group, a substituted or unsubstituted C6 to C30 aryl group and a substituted or unsubstituted C3 to C30 heteroaryl group, each of n1 and n2 is independently 0 or 1, one of X.sub.1 and X.sub.2 is a single bond, the other one of X.sub.1 and X.sub.2 is selected from O, S, Se, NR.sub.25 and C(R.sub.25).sub.2, one of X.sub.3 and X.sub.4 is a single bond, the other one of X.sub.3 and X.sub.4 is selected from O, S, Se, NR.sub.26 and C(R.sub.26).sub.2, and each of R.sub.25 and R.sub.26 is independently selected from the group consisting of hydrogen, deuterium, halogen, cyano, a substituted or unsubstituted C1 to C20 alkyl group, a substituted or unsubstituted C3 to C30 cycloalkyl group, a substituted or unsubstituted C1 to C20 alkoxy group, a substituted or unsubstituted C1 to C20 alkylamino group, a substituted or unsubstituted C6 to C30 arylamino group, a substituted or unsubstituted C1 to C20 alkylsilyl group, a substituted or unsubstituted C6 to C30 arylsilyl group, a substituted or unsubstituted C6 to C30 aryl group and a substituted or unsubstituted C3 to C30 heteroaryl group, in each of Formulas 1-1 and 1-2, the mark "*" denotes a bonding site, wherein the second compound is represented by Formula 3: ##STR00049## wherein in Formula 3, each of b1, b2, b3 and b4 is independently an integer of 0 to 4, when b1 is 2 or more, two or more R.sub.41 are same or different, when b2 is 2 or more, two or more R.sub.42 are same or different, when b3 is 2 or more, two or more R.sub.43 are same or different, when b4 is 2 or more, two or more R.sub.44 are same or different, and each of R.sub.41 to R.sub.44 is independently selected from the group consisting of deuterium, halogen, cyano, a substituted or unsubstituted C1 to C20 alkyl group, a substituted or unsubstituted C1 to C20 alkoxy group, a substituted or unsubstituted C1 to C20 alkylamino group, a substituted or unsubstituted C6 to C30 arylamino group, a substituted or unsubstituted C1 to C20 alkylsilyl group, a substituted or unsubstituted C6 to C30 arylsilyl group, a substituted or unsubstituted C6 to C30 aryl group and a substituted or unsubstituted C3 to C30 heteroaryl group, and wherein the third compound is represented by Formula 5: ##STR00050## wherein in Formula 5, each of d1, d2 and d3 is independently an integer of 0 to 4, d4 is an integer of 0 to 3, d5 is an integer of 0 to 2, when d1 is 2 or more, two or more R.sub.51 are same or different, when d2 is 2 or more, two or more R.sub.52 are same or different, when d3 is 2 or more, two or more R.sub.53 are same or different, when d4 is 2 or more, two or more R.sub.54 are same or different, when d5 is 2, two R.sub.55 are same or different, each of R.sub.51 to R.sub.55 is independently selected from the group consisting of deuterium, halogen, cyano, a substituted or unsubstituted C1 to C20 alkyl group, a substituted or unsubstituted C3 to C30 cycloalkyl group, a substituted or unsubstituted C1 to C20 alkylsilyl group, a substituted or unsubstituted C1 to C20 alkylamino group, a substituted or unsubstituted C6 to C30 arylamino group, a substituted or unsubstituted C6 to C30 arylsilyl group, a substituted or unsubstituted C6 to C30 aryl group and a substituted or unsubstituted C3 to C30 heteroaryl group, and R.sub.56 is selected from the group consisting of hydrogen, deuterium, halogen, cyano, a substituted or unsubstituted C1 to C20 alkyl group, a substituted or unsubstituted C3 to C30 cycloalkyl group, a substituted or unsubstituted C1 to C20 alkylsilyl group, a substituted or unsubstituted C1 to C20 alkylamino group, a substituted or unsubstituted C6 to C30 arylamino group, a substituted or unsubstituted C6 to C30 arylsilyl group, a substituted or unsubstituted C6 to C30 aryl group and a substituted or unsubstituted C3 to C30 heteroaryl group.

2. The organic light emitting diode according to claim 1, wherein Formula 1-1 is represented by Formula 1-1a: ##STR00051## wherein in Formula 1-1a, a6 is an integer of 0 to 4, each of a7, a8 and a9 is independently an integer of 0 to 5, when a6 is 2 or more, two or more R.sub.31 are same or different, when a7 is 2 or more, two or more R.sub.32 are same or different, when a8 is 2 or more, two or more R.sub.33 are same or different, when a9 is 2 or more, two or more R.sub.34 are same or different, and each of R.sub.31, R.sub.32, R.sub.33 and R.sub.34 is independently selected from the group consisting of deuterium, halogen, cyano, a substituted or unsubstituted C1 to C20 alkyl group, a substituted or unsubstituted C3 to C30 cycloalkyl group, a substituted or unsubstituted C1 to C20 alkoxy group, a substituted or unsubstituted C1 to C20 alkylamino group, a substituted or unsubstituted C6 to C30 arylamino group, a substituted or unsubstituted C1 to C20 alkylsilyl group, a substituted or unsubstituted C6 to C30 arylsilyl group, a substituted or unsubstituted C6 to C30 aryl group and a substituted or unsubstituted C3 to C30 heteroaryl group.

3. The organic light emitting diode according to claim 1, wherein the first compound is one of compounds in Formula 2: ##STR00052## ##STR00053## ##STR00054## ##STR00055## ##STR00056## ##STR00057## ##STR00058## ##STR00059## ##STR00060## ##STR00061## ##STR00062## ##STR00063## ##STR00064## ##STR00065## ##STR00066## ##STR00067## ##STR00068## ##STR00069## ##STR00070## ##STR00071## ##STR00072##

4. The organic light emitting diode according to claim 1, wherein the second compound is one of compounds in Formula 4: ##STR00073## ##STR00074## ##STR00075##

5. The organic light emitting diode according to claim 1, wherein the third compound is one of compounds in Formula 6: ##STR00076## ##STR00077## ##STR00078##

6. The organic light emitting diode according to claim 1, wherein a lowest unoccupied molecular orbital (LUMO) energy level of the first compound is lower than a LUMO energy level of the second compound and is higher than a LUMO energy level of the third compound, and wherein a difference between the LUMO energy level of the first compound and the LUMO energy level of the third compound is smaller than a difference between the LUMO energy level of the first compound and the LUMO energy level of the second compound.

7. The organic light emitting diode according to claim 6, wherein a difference between a HOMO energy level of the first compound and a HOMO energy level of the third compound is 0.1 eV or more and 0.3 eV or less.

8. The organic light emitting diode according to claim 7, wherein the HOMO energy level of the first compound is higher than a HOMO energy level of the second compound and is lower than the HOMO energy level of the third compound, and wherein a difference between the HOMO energy level of the first compound and the HOMO energy level of the third compound is smaller than a difference between the HOMO energy level of the first compound and the HOMO energy level of the second compound.

9. The organic light emitting diode according to claim 1, wherein the first blue emitting material layer further includes a first fluorescent compound represented by Formula 7: ##STR00079## wherein in Formula 7, each of e1 and e6 is independently an integer of 0 to 4, each of e2 to e5 is independently an integer of 0 to 5, when e1 is 2 or more, two or more R.sub.61 are same or

different, when e2 is 2 or more, two or more R.sub.62 are same or different, when e3 is 2 or more, two or more R.sub.63 are same or different, when e4 is 2 or more, two or more R.sub.64 are same or different, when e5 is 2 or more, two or more R.sub.65 are same or different, when e6 is 2 or more, two or more R.sub.66 are same or different, each of R.sub.61 to R.sub.66 is independently selected from the group consisting of deuterium, a substituted or unsubstituted C1 to C20 alkyl group, a substituted or unsubstituted C1 to C20 alkoxy group, a substituted or unsubstituted C6 to C30 arylsilyl group, a substituted or unsubstituted C6 to C30 arylamino group, a substituted or unsubstituted C6 to C30 aryl group and a substituted or unsubstituted C3 to C30 heteroaryl group, and each of Ar.sub.4 and Ar.sub.5 is independently selected from the group consisting of a substituted or unsubstituted C6 to C30 arylamino group, a substituted or unsubstituted C6 to C30 aryl group and a substituted or unsubstituted C3 to C30 heteroaryl group.
10. The organic light emitting diode according to claim 9, wherein the first fluorescent compound is one of compounds in Formula 8: ##STR00080## ##STR00081##
11. The organic light emitting diode according to claim 9, wherein each of a difference between a singlet energy level and a triplet energy level in the first compound and a difference between a singlet energy level and a triplet energy level in the first fluorescent compound is 0.3 eV or less.
12. The organic light emitting diode according to claim 11, wherein the difference between a singlet energy level and a triplet energy level in the first fluorescent compound is smaller than the difference between a singlet energy level and a triplet energy level in the first compound.
13. The organic light emitting diode according to claim 9, wherein a lowest unoccupied molecular orbital (LUMO) energy level of the first compound is lower than a LUMO energy level of the second compound and is higher than a LUMO energy level of the third compound, wherein the LUMO energy level of the third compound is higher than a LUMO energy level of the first fluorescent compound, wherein a difference between the LUMO energy level of the first compound and the LUMO energy level of the third compound is smaller than a difference between the LUMO energy level of the first compound and the LUMO energy level of the second compound, and wherein a difference between the LUMO energy level of the third compound and the LUMO energy level of the first fluorescent compound is equal to or greater than a difference between the LUMO energy level of the first compound and the LUMO energy level of the second compound.
14. The organic light emitting diode according to claim 1, further comprising: a second blue emitting material layer positioned between the first blue emitting material layer and the second electrode.
15. The organic light emitting diode according to claim 14, wherein the second blue emitting material layer includes a fourth compound, a fifth compound and a sixth compound, and wherein the fourth compound is represented by Formula 1, the fifth compound is represented by Formula 3, and the sixth compound is represented by Formula 5.
16. The organic light emitting diode according to claim 15, wherein the second blue emitting material layer further includes a second fluorescent compound represented by Formula 7: ##STR00082## wherein in Formula 7, each of e1 and e6 is independently an integer of 0 to 4, each of e2 to e5 is independently an integer of 0 to 5, when e1 is 2 or more, two or more R.sub.61 are same or different, when e2 is 2 or more, two or more R.sub.62 are same or different, when e3 is 2 or more, two or more R.sub.63 are same or different, when e4 is 2 or more, two or more R.sub.64 are same or different, when e5 is 2 or more, two or more R.sub.65 are same or different, when e6 is 2 or more, two or more R.sub.66 are same or different, each of R.sub.61 to R.sub.66 is independently selected from the group consisting of deuterium, a substituted or unsubstituted C1 to C20 alkyl group, a substituted or unsubstituted C1 to C20 alkoxy group, a substituted or unsubstituted C6 to C30 arylsilyl group, a substituted or unsubstituted C6 to C30 arylamino group, a substituted or unsubstituted C6 to C30 aryl group and a substituted or unsubstituted C3 to C30 heteroaryl group, and each of Ar.sub.4 and Ar.sub.5 is independently selected from the group consisting of a substituted or unsubstituted C6 to C30 arylamino group, a substituted or

unsubstituted C6 to C30 aryl group and a substituted or unsubstituted C3 to C30 heteroaryl group.
17. The organic light emitting diode according to claim 15, further comprising: a third emitting material layer positioned between the first and second blue emitting material layers and including a red emitting material layer, a yellow-green emitting material layer and a green emitting material layer.
18. An organic light emitting device, comprising: a substrate; the organic light emitting diode according to claim 1 disposed on the substrate; and an encapsulation layer covering the organic light emitting diode.
19. The organic light emitting device according to claim 18, wherein the first compound is one of compounds in Formula 2: ##STR00083## ##STR00084## ##STR00085## ##STR00086## ##STR00087## ##STR00088## ##STR00089## ##STR00090## ##STR00091## ##STR00092## ##STR00093## ##STR00094## ##STR00095## ##STR00096## ##STR00097## ##STR00098## ##STR00099## ##STR00100## ##STR00101## ##STR00102## ##STR00103## ##STR00104##
20. The organic light emitting device according to claim 18, wherein the second compound is one of compounds in Formula 4: ##STR00105## ##STR00106## ##STR00107##
21. The organic light emitting device according to claim 18, wherein the third compound is one of compounds in Formula 6: ##STR00108## ##STR00109## ##STR00110##